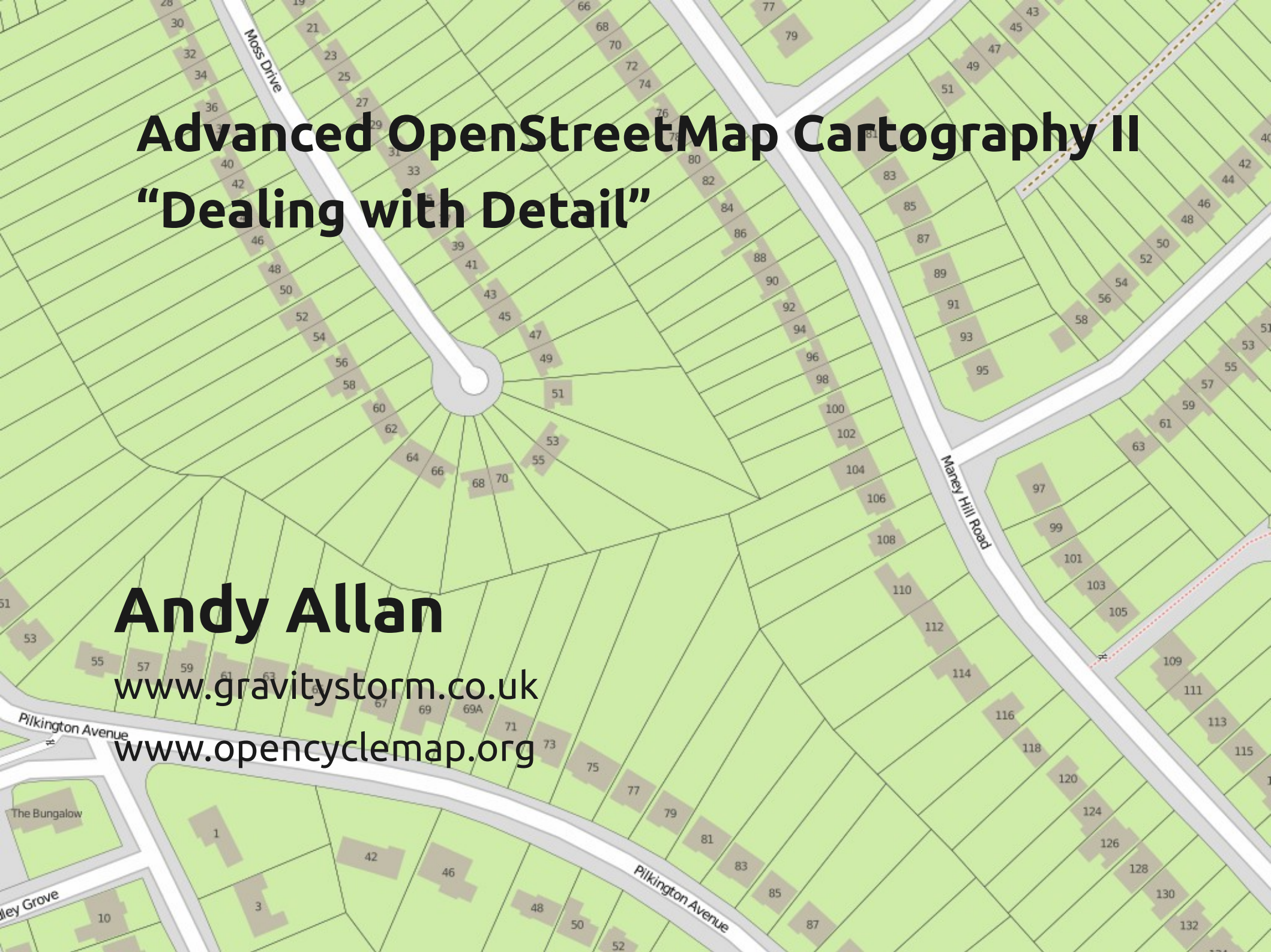


Advanced OpenStreetMap Cartography II

“Dealing with Detail”

Andy Allan

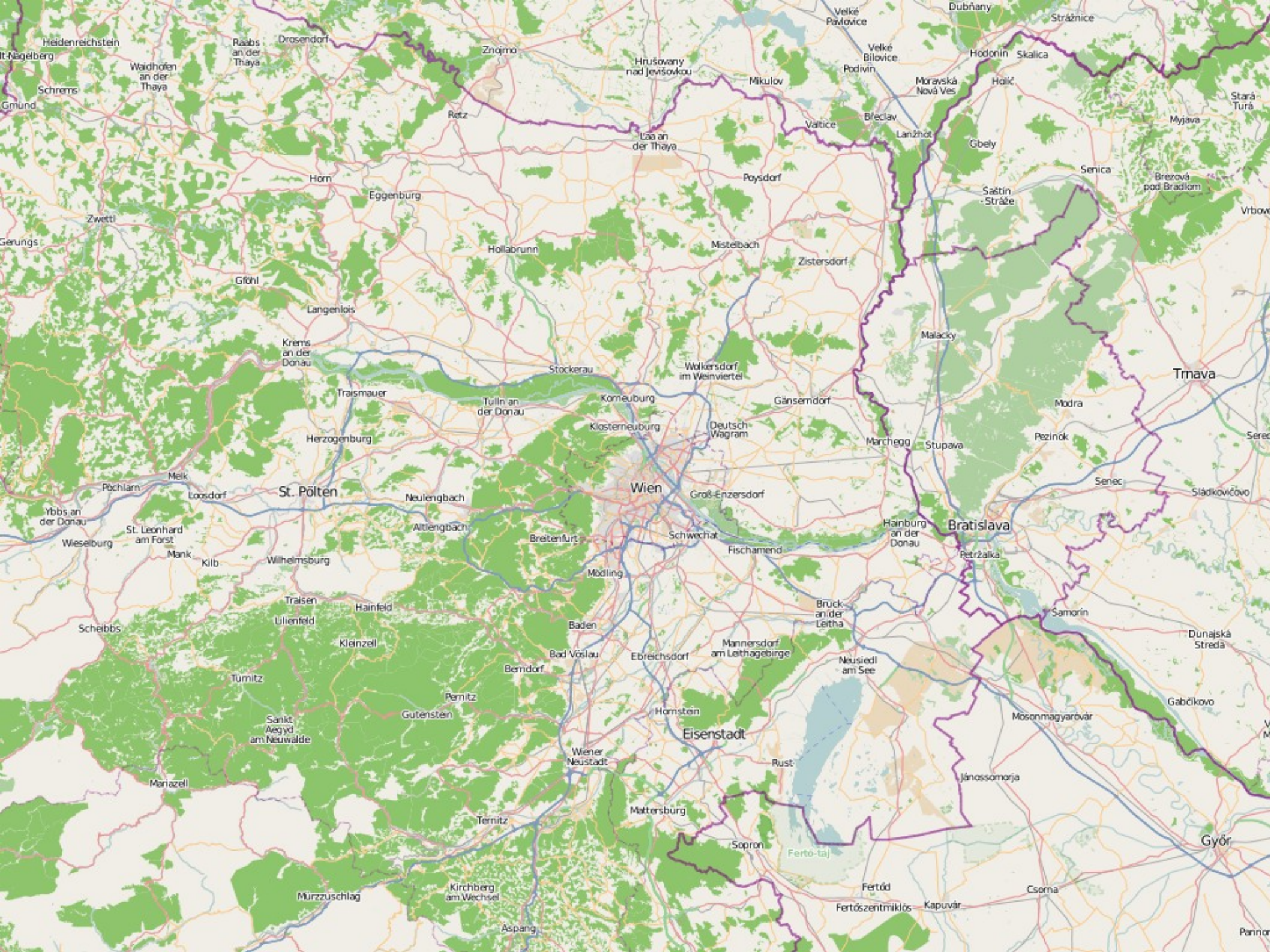
www.gravitystorm.co.uk

www.opencyclemap.org

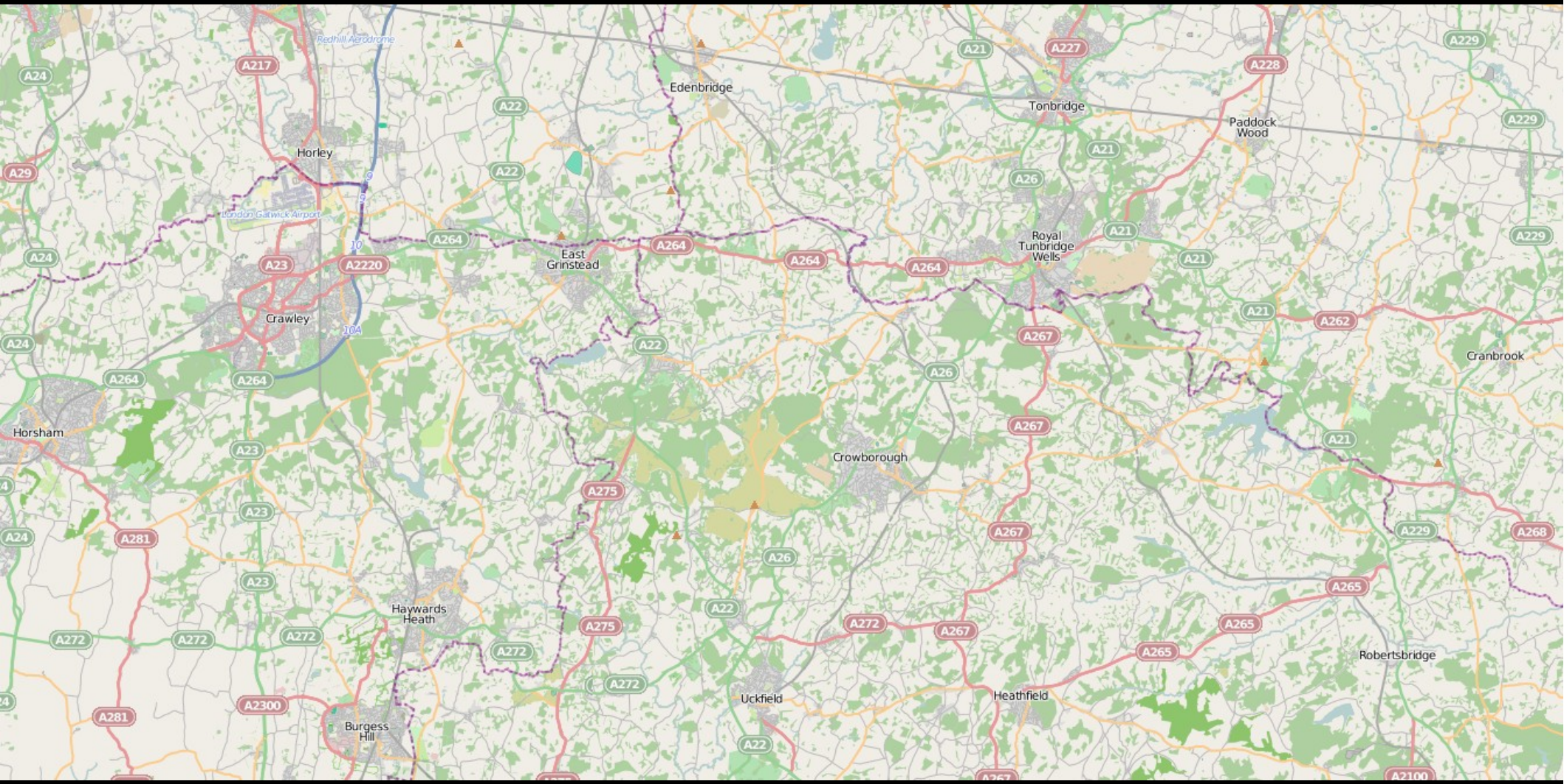

Moss Drive
Maney Hill Road
Pilkington Avenue
Pilkington Avenue
The Bungalow
ley Grove

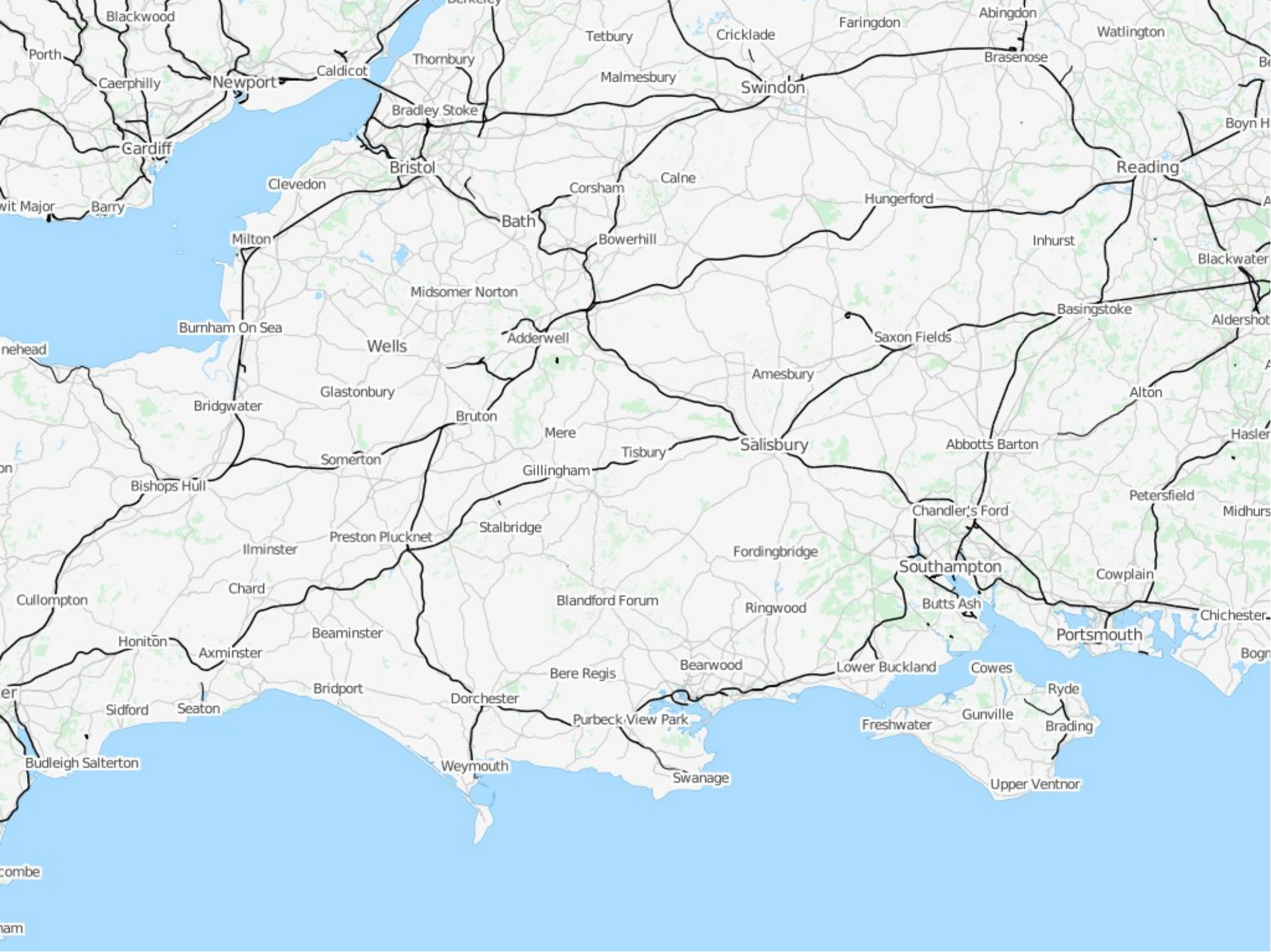
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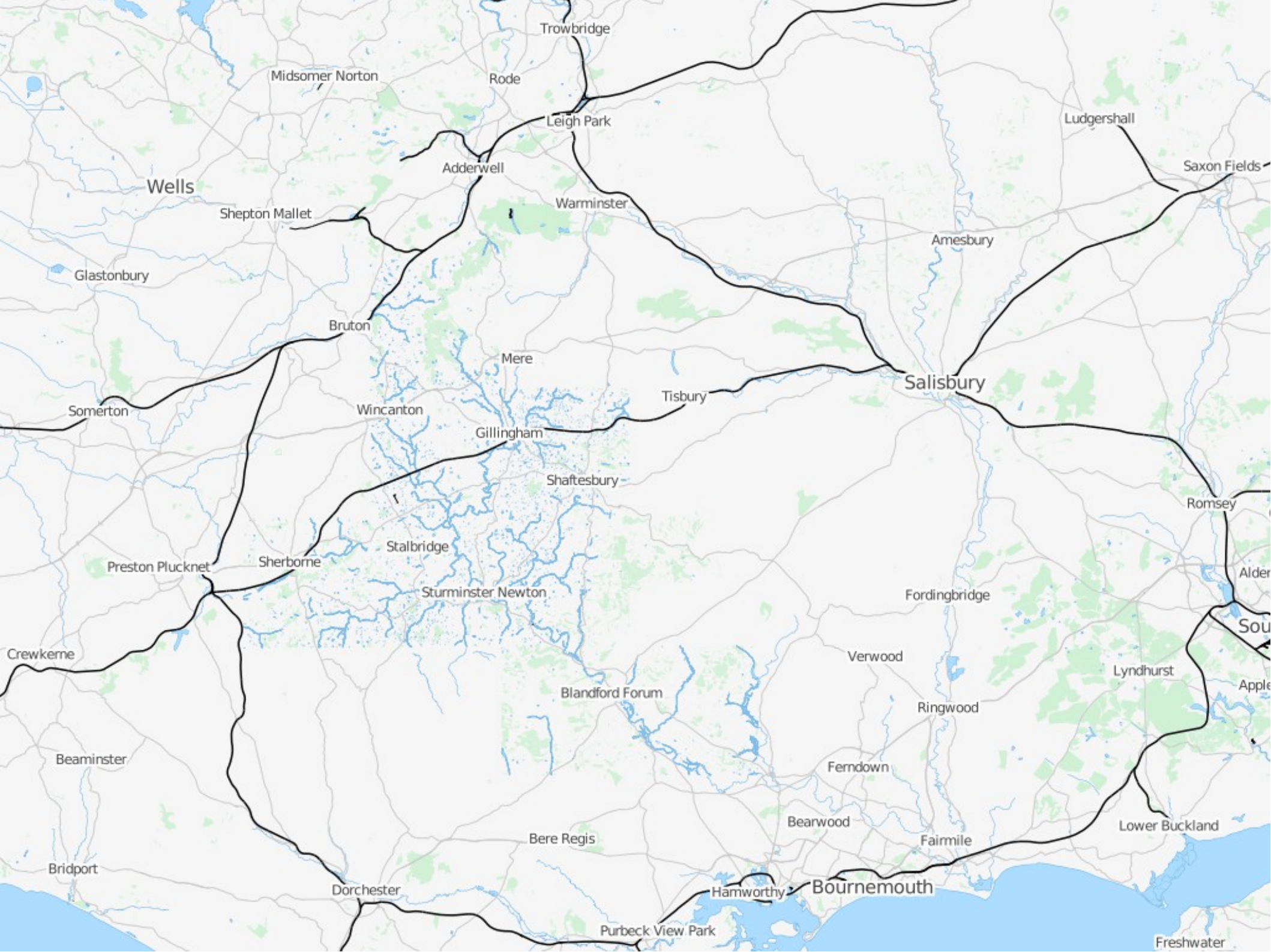

Burggarten
Kunsthistorisches Museum
Museumsquartier
Burgring
Babenbergerstraße
Nibelungengasse
Operngasse
Goethegasse
Staatsoper
Annagasse
Krugerstraße
Fichtegasse
Johannesgasse
BORG für Musik und Kunst
Walfischgasse
Hegelgasse
Schubertring
Kärntner Ring / Oper
Mahlerstraße
Stadtpark
Kärntner Straße
Kärntner Ring
Schwarzenbergplatz
Lothringerstraße
Am Heumarkt
Elisabethstraße
Schillerplatz
Schillerpark
Akademie der bildenden Künste
Makartgasse
GRG Rahlgasse
Getreidemarkt
Theobaldgasse
Technische Universität Wien
Audimax
Hoftrakt
Bösendorferstraße
Karlsplatz
Canovagasse
Resselpark
Treitlstraße
Wiedner Hauptstraße
Millöckergasse
Lehargasse
Mariahilfer Straße
Girardigasse
Laimgrubengasse
Faulmannsgasse
Resselgasse
Kartsplatz
Brucknerstraße
Maderstraße
Zaunergasse
Liststraße
Traungasse
Marokkanergasse
Ölzeltgasse
Lagergasse
Salesianergasse
Rechte Wienzeile
Naschmarkt
Lili's Market (Chinese)
Hofer
Billa
Spar
Spar Gourmet
Margaretenstraße
Mühlgasse
Heumühlgasse
Preßgasse
Kettenbrückengasse
Paniglgasse
Frankenberggasse
Kartsgasse
Gußhausstraße
Tanzschule Dorner
Schwindgasse
Wohllebengasse
Palais Schwarzenberg
Unteres Belvedere
Kammergarten
Paulanergasse
Zielpunkt
Waaggasse
Floragasse
Taubstummengasse
Argentinierstraße
Plößlgasse
Schönbrunner Straße
Franzensgasse
Große Neugasse
Rienößlgasse
Freundgasse
Henrühlgasse
Weihburggasse
Stadtpark
Johannesgasse
Am Heumarkt
Lothringerstraße
Schwarzenbergplatz
Gußhausstraße
Wiedner Hauptstraße
Taubstummengasse
Plößlgasse
U1
U2
U4
2,D
71


Heidenreichstein
Raabs an der Thaya
Drosendorf
Znojmo
Velké Pavlovice
Dubňany
Strážnice
Hodonín
Skalica
Velké Bílovice
Podivín
Hrušovany nad Jevišovkou
Mikulov
Moravská Nová Ves
Holíč
Waidhofen an der Thaya
Schrems
Gmünd
Retz
Laa an der Thaya
Valtice
Břeclav
Lanžhot
Gbely
Stará Turá
Myjava
Horn
Eggenburg
Poysdorf
Senica
Šaštín - Stráže
Brezová pod Bradlom
Zwettl
Gerungs
Hollabrunn
Mistelbach
Zistersdorf
Gföhl
Langenlois
Malacky
Krems an der Donau
Stockerau
Wolkersdorf im Weinviertel
Trnava
Traismauer
Tulln an der Donau
Korneuburg
Gänserndorf
Modra
Klosterneuburg
Deutsch Wagram
Herzogenburg
Marchegg
Stupava
Pezinok
Melk
Pöchlarn
St. Pölten
Loosdorf
Neulengbach
Wien
Senec
Sládkovičovo
Groß-Enzersdorf
Ybbs an der Donau
Altlengbach
Hainburg an der Donau
Bratislava
St. Leonhard am Forst
Schwechat
Wieselburg
Breitenfurt
Fischamend
Petržalka
Mank
Kilb
Wilhelmsburg
Modling
Traisen
Hainfeld
Brück an der Leitha
Samorín
Lilienfeld
Scheibbs
Baden
Dunajská Streda
Kleinzell
Bad Vöslau
Ebreichsdorf
Mannersdorf am Leithagebirge
Neusiedl am See
Bernhof
Türnitz
Pernitz
Gutenstein
Gabčíkovo
Hornstein
Mosonmagyaróvár
Sankt Aegyd am Neuwalde
Eisenstadt
Wiener Neustadt
Rust
Mariazell
Jánossomorja
Mattersburg
Ternitz
Sopron
Győr
Fertő-táj
Kirchberg am Wechsel
Fertőd
Csorna
Mürzzuschlag
Fertőszentmiklós
Kapuvár
Aspang

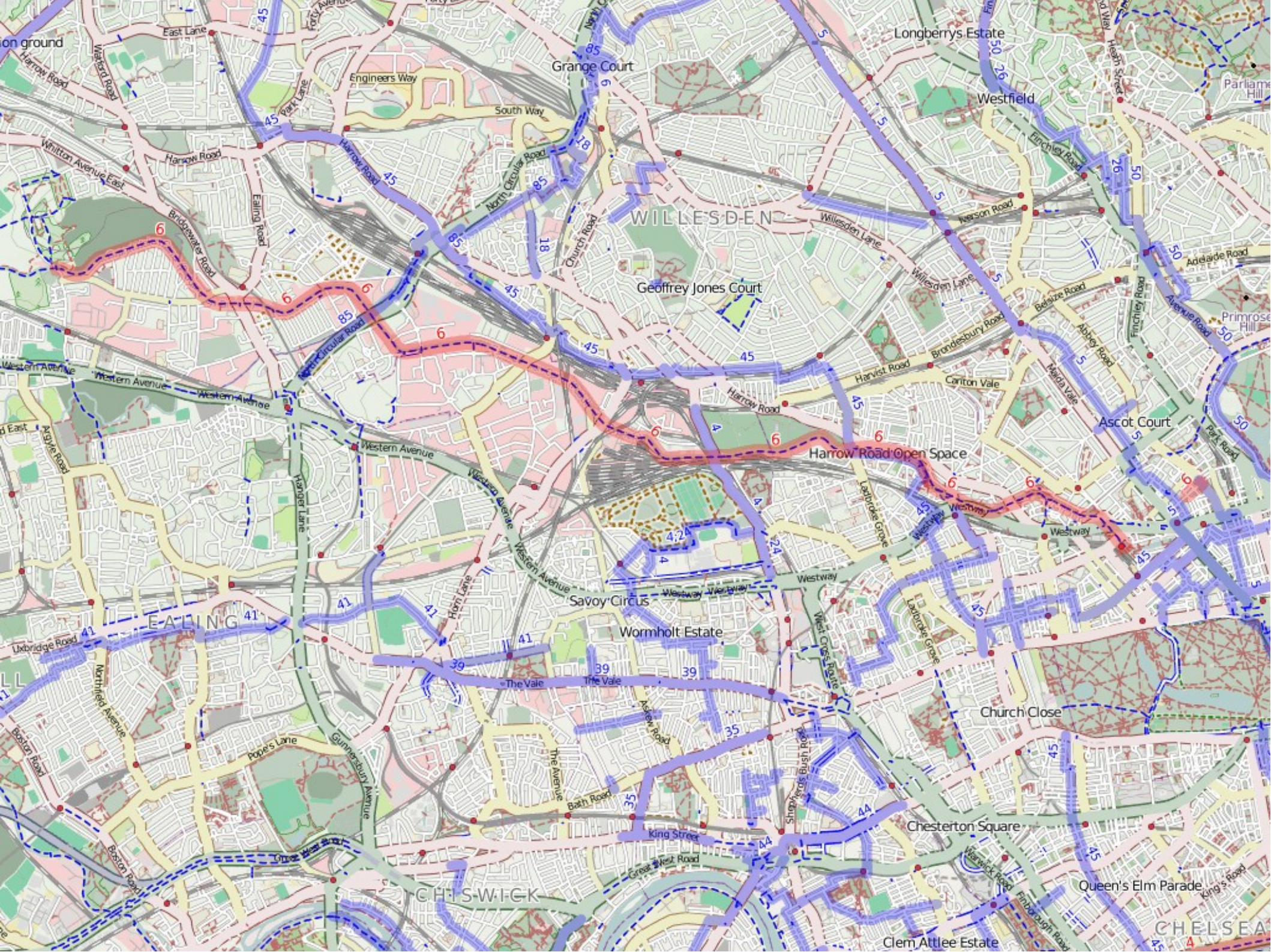
Small Polygons

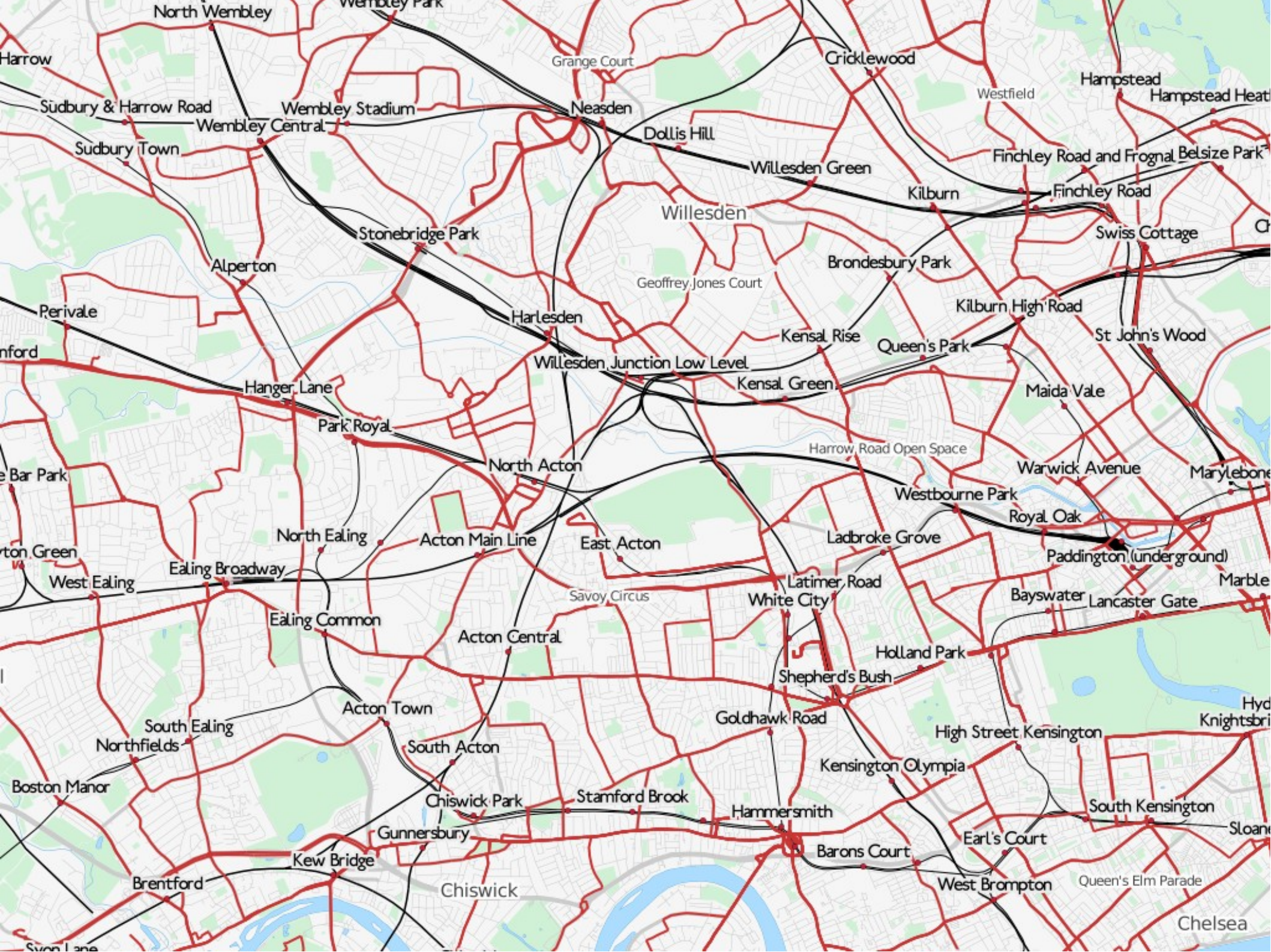

Redhill Aerodrome
Edenbridge
Tonbridge
Paddock Wood
Horley
London Gatwick Airport
East Grinstead
Royal Tunbridge Wells
Crawley
Cranbrook
Horsham
Crowborough
Haywards Heath
Robertsbridge
Uckfield
Heathfield
Burgess Hill
A217
A227
A228
A229
A24
A22
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A2220
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A267
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A268
A2300
A2100

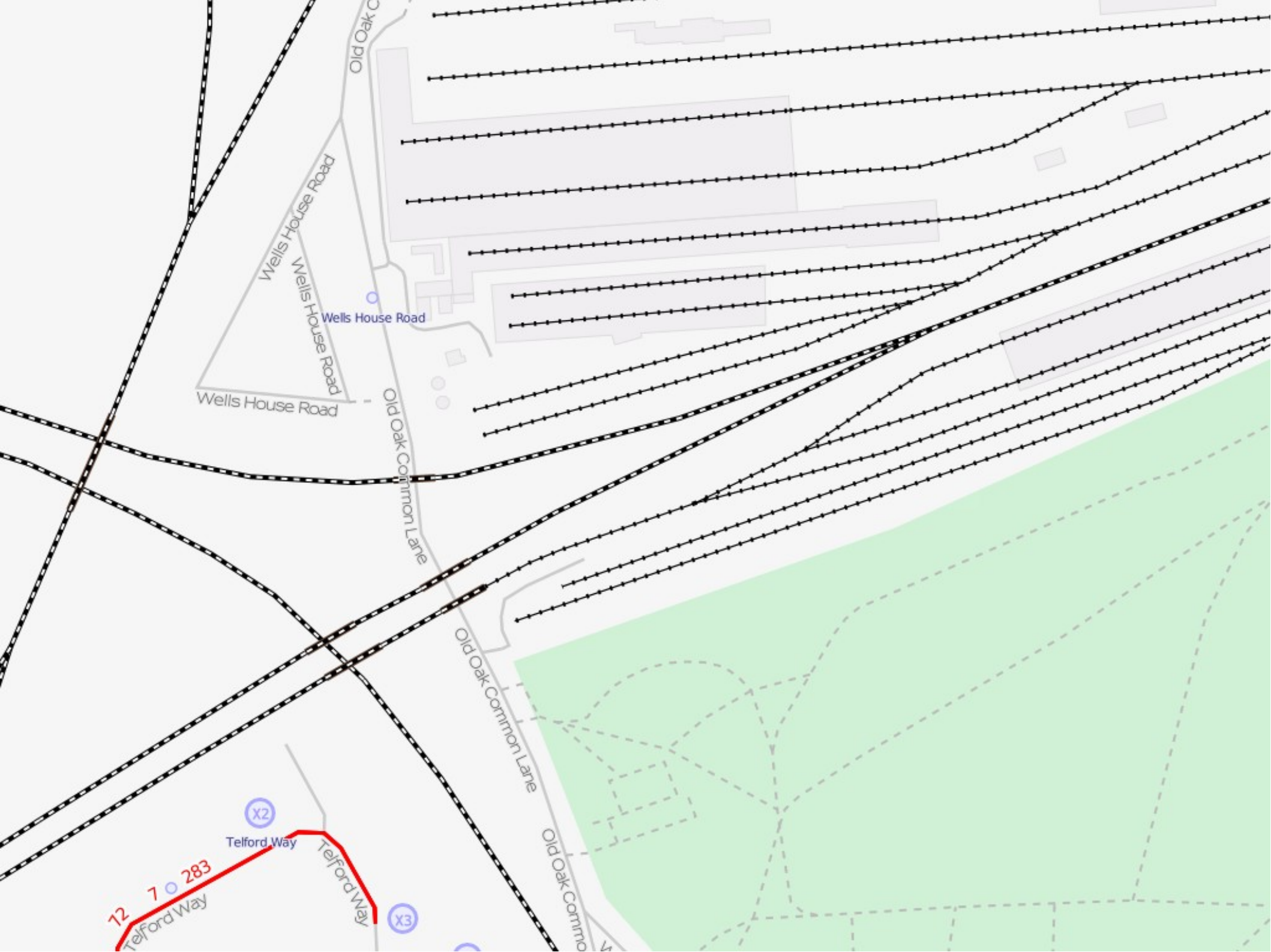

Blackwood
Porth
Caerphilly
Newport
Caldicot
Thornbury
Tetbury
Cricklade
Faringdon
Abingdon
Watlington
Brasenose
Malmesbury
Swindon
Bradley Stoke
Gardiff
Bristol
Clevedon
Barry
Corsham
Calne
Hungerford
Reading
Bath
Milton
Bowerhill
Inhurst
Blackwater
Midsomer Norton
Basingstoke
Burnham On Sea
Wells
Adderwell
Saxon Fields
Aldershot
Amesbury
Glastonbury
Alton
Bridgwater
Bruton
Mere
Salisbury
Abbotts Barton
Tisbury
Somerton
Gillingham
Bishops Hull
Petersfield
Chandler's Ford
Stalbridge
Preston Plucknet
Ilminster
Fordingbridge
Southampton
Cowplain
Chard
Cullompton
Blandford Forum
Ringwood
Butts Ash
Chichester
Beaminster
Portsmouth
Honiton
Axminster
Bearwood
Lower Buckland
Cowes
Bere Regis
Ryde
Bridport
Dorchester
Sidford
Seaton
Purbeck View Park
Gunville
Freshwater
Brading
Budleigh Salterton
Weymouth
Swanage
Upper Ventnor


Trowbridge
Midsomer Norton
Rode
Leigh Park
Ludgershall
Adderwell
Saxon Fields
Wells
Warminster
Shepton Mallet
Amesbury
Glastonbury
Bruton
Mere
Salisbury
Tisbury
Somerton
Wincanton
Gillingham
Shaftesbury
Romsey
Stalbridge
Preston Plucknet
Sherborne
Sturminster Newton
Fordingbridge
Crewkerne
Verwood
Lyndhurst
Blandford Forum
Ringwood
Beaminster
Ferndown
Bearwood
Lower Buckland
Bere Regis
Fairmile
Bridport
Dorchester
Hamworthy
Bournemouth
Purbeck View Park
Freshwater

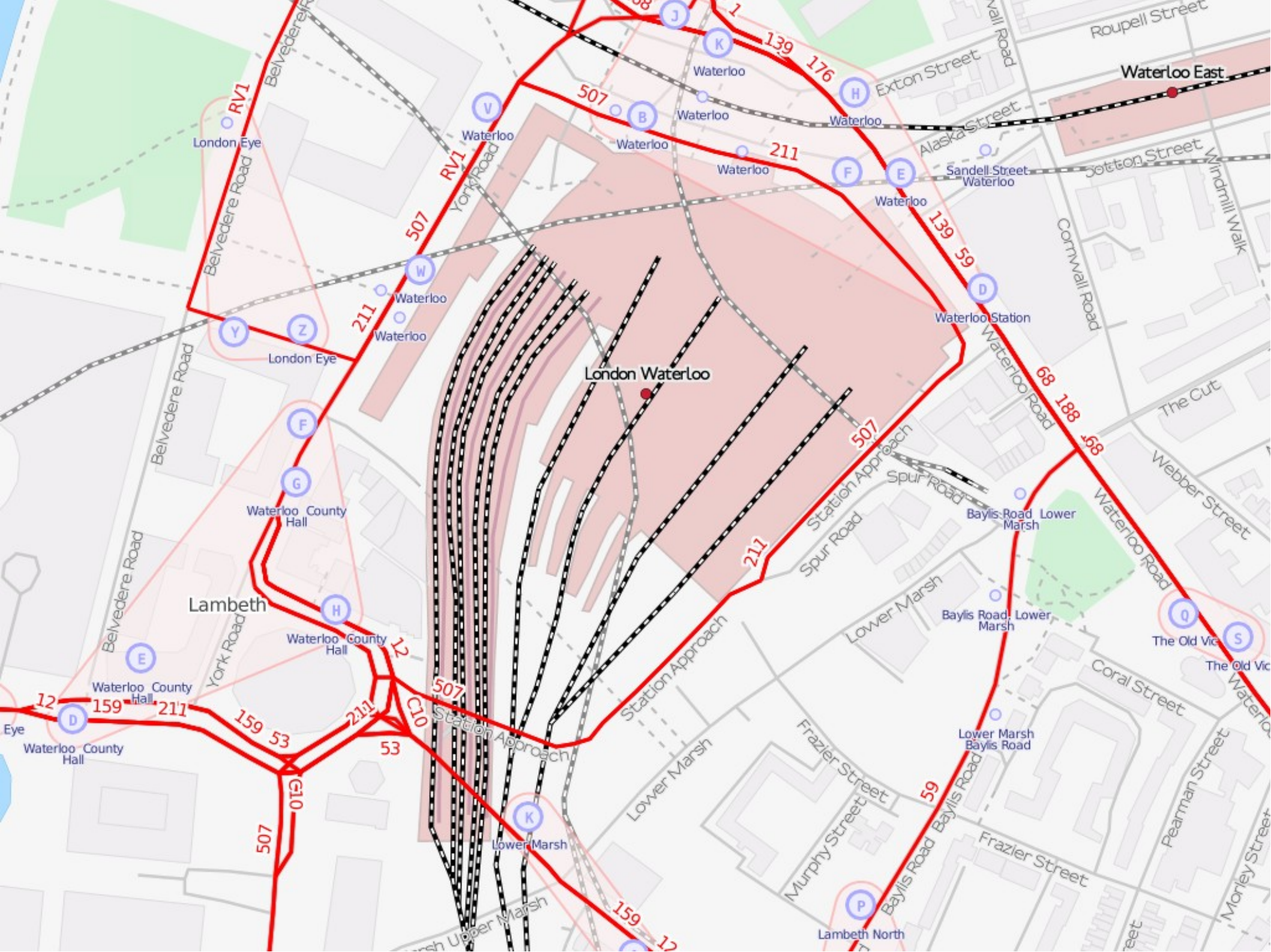
Extra detail for better rendering

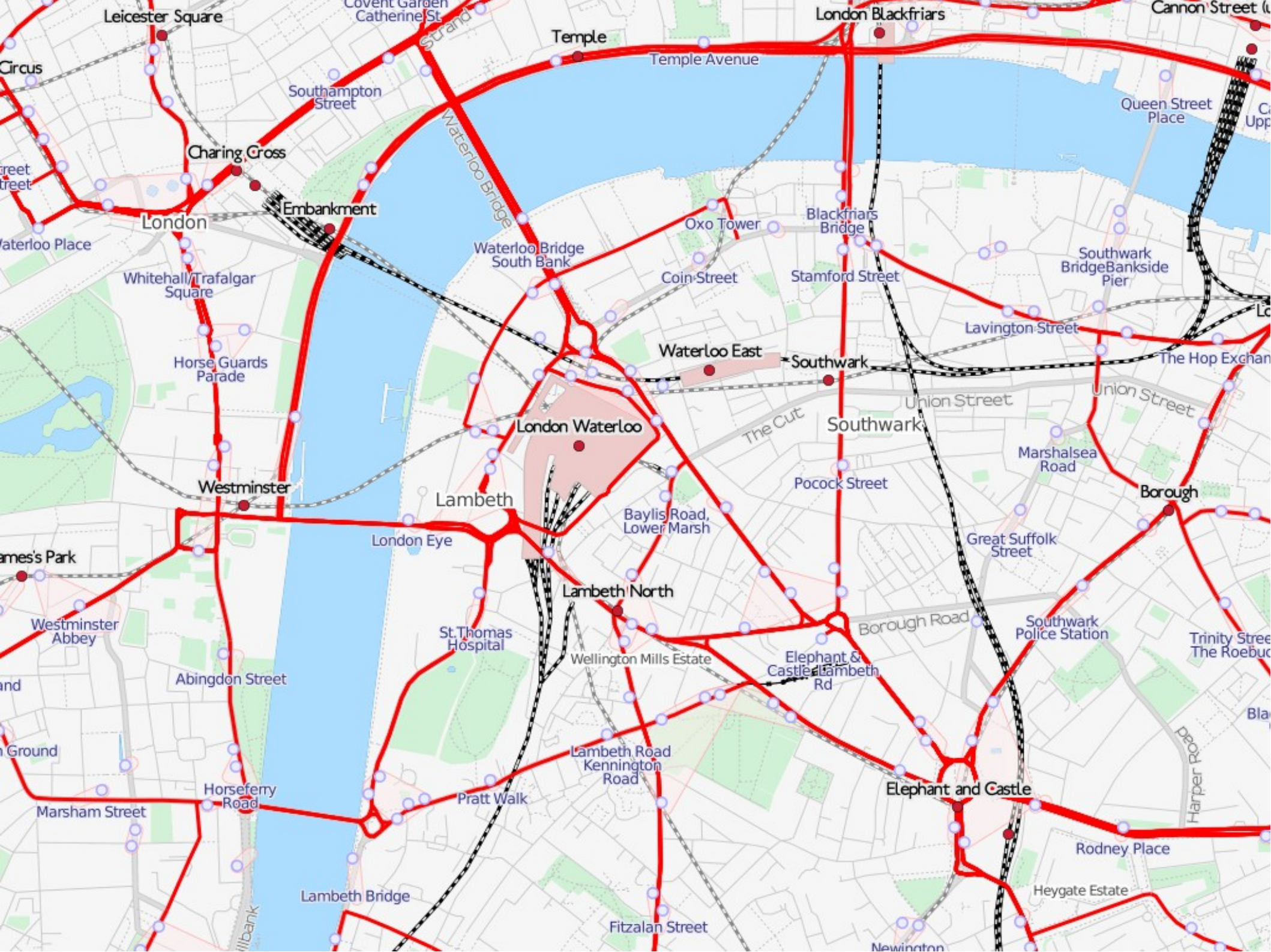

Grange Court
Longberrys Estate
Westfield
WILLESDEN
Geoffrey Jones Court
Harrow Road Open Space
Ascot Court
Savoy Circus
Wormholt Estate
EALING
Church Close
Chesterton Square
Queen's Elm Parade
CHISWICK
Clem Attlee Estate
CHELSEA
Harrow Road
Western Avenue
North Circular Road
Westway
The Vale
King Street
Bath Road
Great West Road
Uxbridge Road
Hanger Lane
Ealing Road
Willesden Lane
Kilburn High Road
Finchley Road
Abbey Road
Ladbroke Grove
Shepherd's Bush Road
Warwick Road
Finborough Road
Pope's Lane
Gunnersbury Avenue
Boston Road
Northfield Avenue
The Avenue
Acton Road
Carlton Vale
Brondesbury Road
Harvist Road
Adelaide Road
Primrose Hill
Parliament Hill
Engineers Way
South Way
Harrow Road
Whitton Avenue East
Bridgewater Road
Park Lane
East Lane
Warkford Road
Forty Avenue
Church Road
Neasden Lane
Belsize Road
Avenue Road
Park Road
Maida Vale
Horn Lane
West Cross Route
Heath Street
Finchley Road


North Wembley
Wembley Park
Harrow
Grange Court
Cricklewood
Hampstead
Westfield
Hampstead Heath
Sudbury & Harrow Road
Wembley Stadium
Neasden
Wembley Central
Dollis Hill
Sudbury Town
Willesden Green
Finchley Road and Frognal
Belsize Park
Kilburn
Finchley Road
Willesden
Stonebridge Park
Swiss Cottage
Alperton
Brondesbury Park
Geoffrey Jones Court
Perivale
Kilburn High Road
Harlesden
Kensal Rise
Queen's Park
St John's Wood
Willesden Junction Low Level
Kensal Green
Hanger Lane
Maida Vale
Park Royal
Harrow Road Open Space
North Acton
Warwick Avenue
Westbourne Park
Royal Oak
North Ealing
Acton Main Line
East Acton
Ladbroke Grove
Paddington (underground)
Ealing Broadway
West Ealing
Latimer Road
Savoy Circus
White City
Bayswater
Lancaster Gate
Ealing Common
Acton Central
Holland Park
Shepherd's Bush
Acton Town
Goldhawk Road
South Ealing
High Street Kensington
Northfields
South Acton
Kensington Olympia
Boston Manor
Chiswick Park
Stamford Brook
Hammersmith
South Kensington
Gunnersbury
Earl's Court
Kew Bridge
Barons Court
Brentford
West Brompton
Queen's Elm Parade
Chiswick
Chelsea

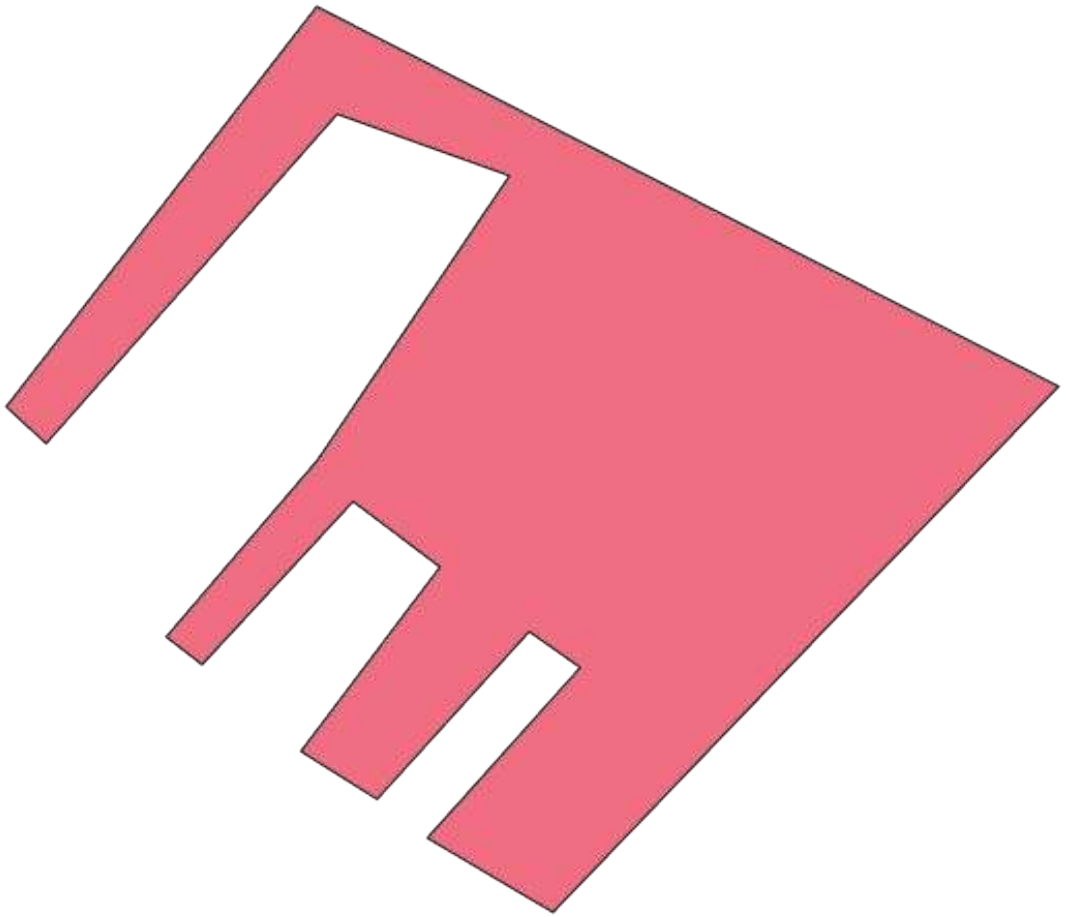

Wells House Road
Wells House Road
Wells House Road
Wells House Road
Old Oak Common Lane
Old Oak Common Lane
Old Oak Common Lane
X2
Telford Way
Telford Way
Telford Way
X3
12
7
283

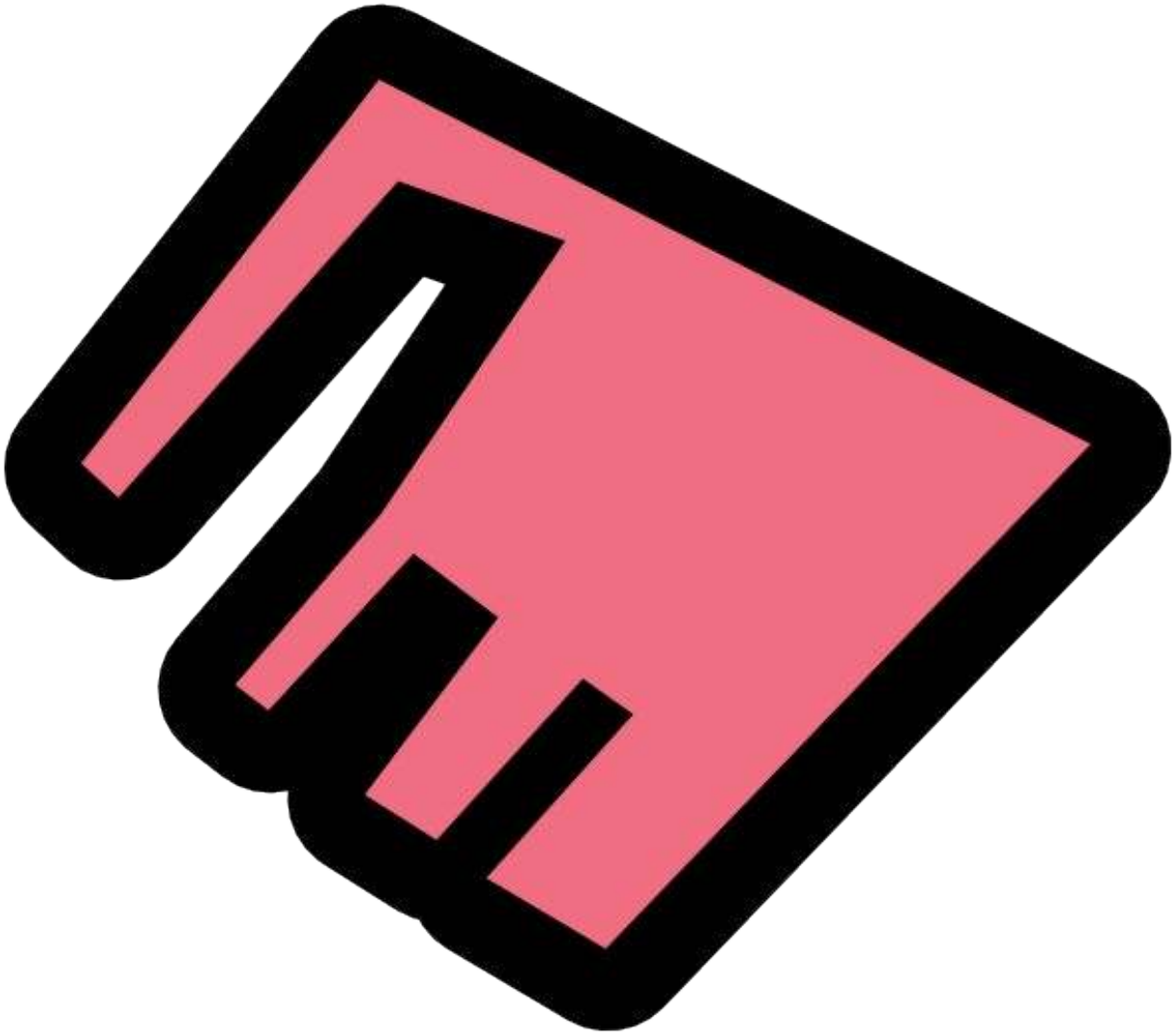
Buffering and Suchlike


London Waterloo
Waterloo East
Lambeth
Waterloo Station
Waterloo
Waterloo Road
Belvedere Road
York Road
Station Approach
Spur Road
Lower Marsh
Baylis Road
Exton Street
Alaska Street
Sandell Street
Jotton Street
Roupell Street
Cornwall Road
Windmill Walk
The Cut
Webber Street
Coral Street
Frazier Street
Murphy Street
Pearman Street
Morley Street
Lambeth North
Lower Marsh Baylis Road
Baylis Road Lower Marsh
Waterloo County Hall
London Eye
The Old Vic
RV1
507
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C10
G10

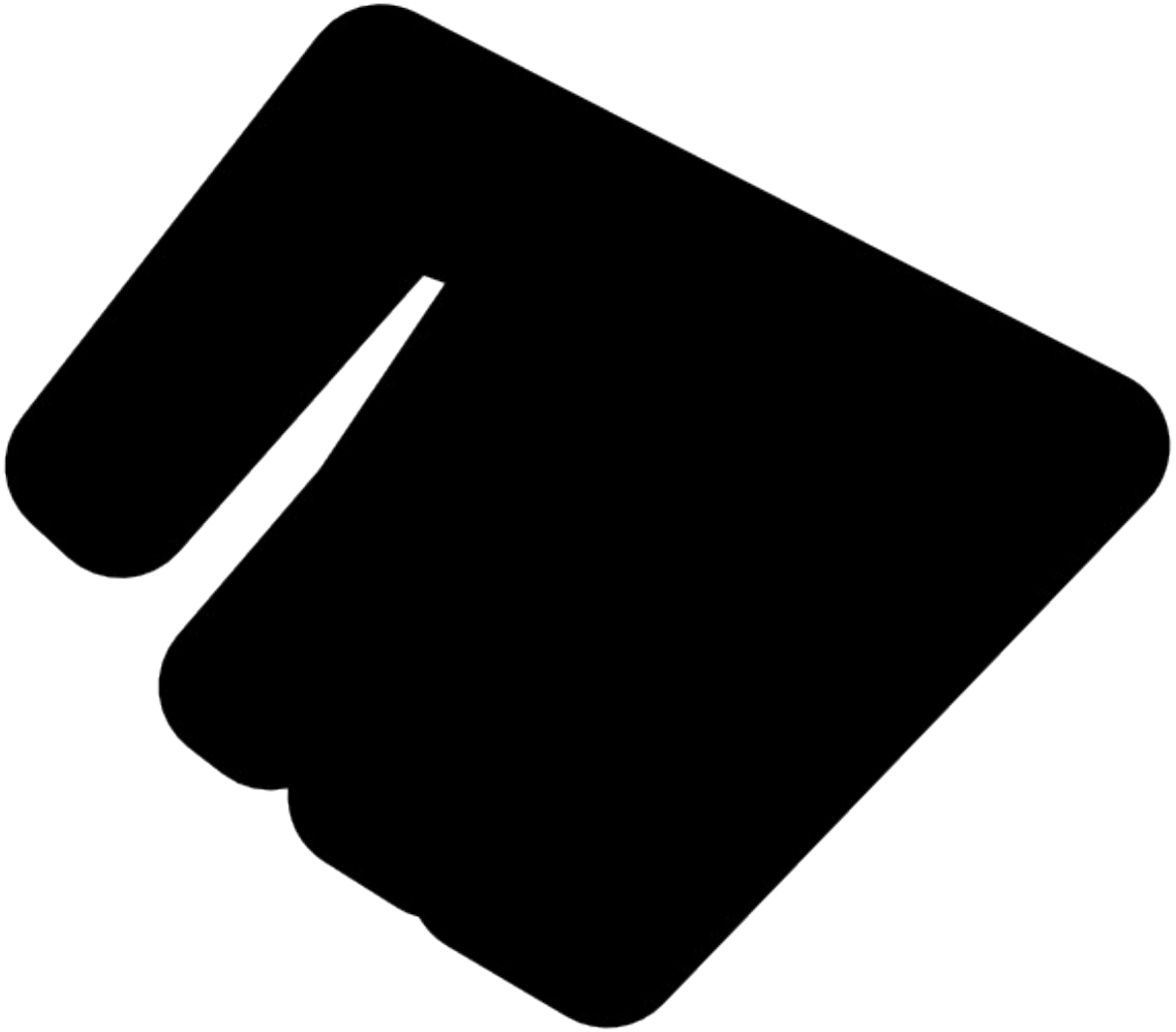

Leicester Square
Covent Garden
Catherine St
Strand
Temple
Temple Avenue
London Blackfriars
Cannon Street
Circus
Southampton Street
Queen Street Place
Charing Cross
Waterloo Bridge
Embankment
London
Oxo Tower
Blackfriars Bridge
Waterloo Bridge South Bank
Coin Street
Stamford Street
Southwark Bridge
Bankside Pier
Whitehall
Trafalgar Square
Lavington Street
Waterloo East
Southwark
Horse Guards Parade
Union Street
The Cut
London Waterloo
Southwark
Marshalsea Road
Westminster
Lambeth
Pocock Street
Borough
Baylis Road, Lower Marsh
London Eye
Great Suffolk Street
Lambeth North
Westminster Abbey
St Thomas Hospital
Borough Road
Southwark Police Station
Wellington Mills Estate
Elephant & Castle
Lambeth Rd
Abingdon Street
Lambeth Road
Kennington Road
Horseferry Road
Pratt Walk
Elephant and Castle
Harper Road
Marsham Street
Rodney Place
Lambeth Bridge
Heygate Estate
Fitzalan Street

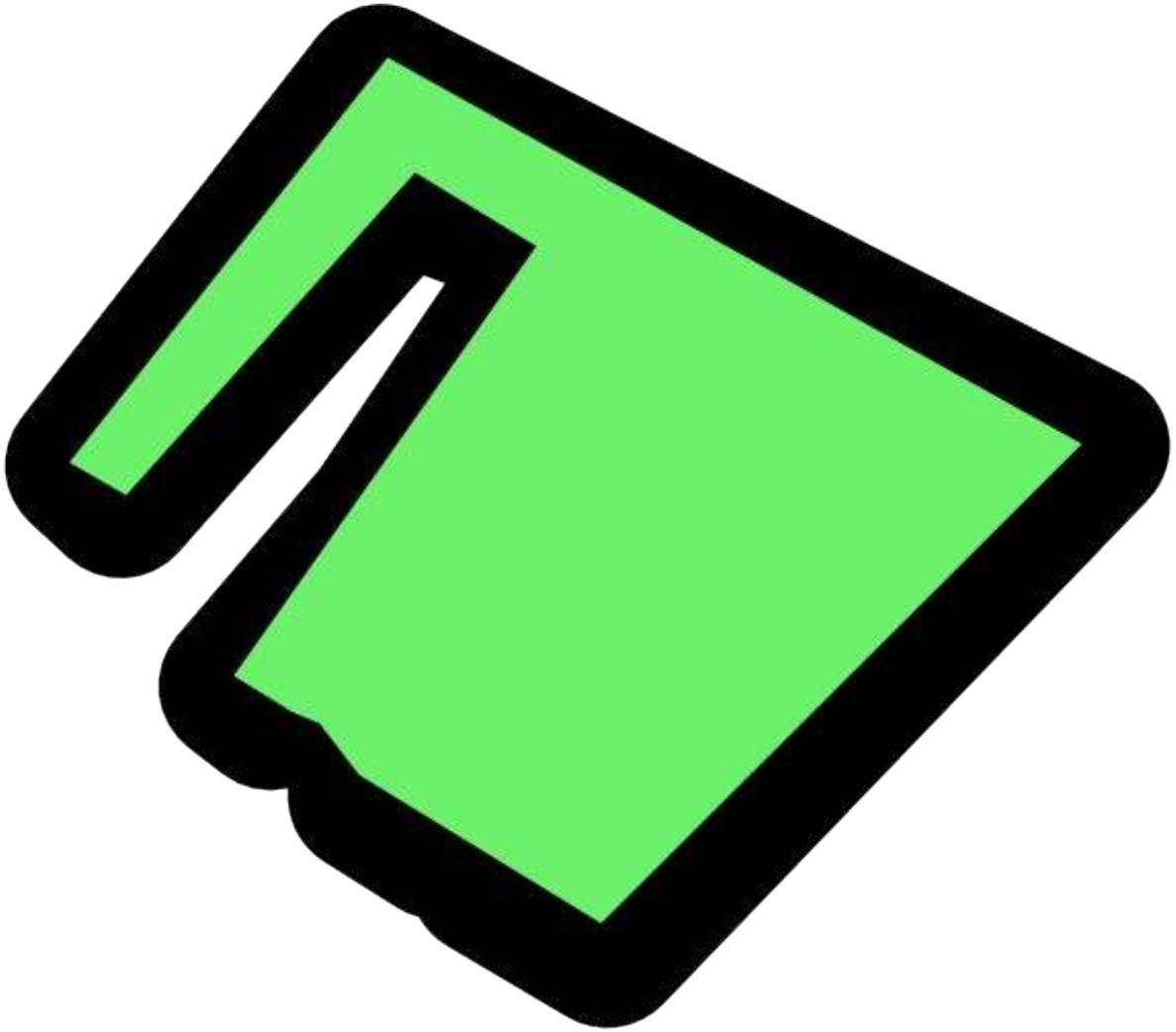
SELECT **way** FROM planet_osm_polygon WHERE building = 'train_station'



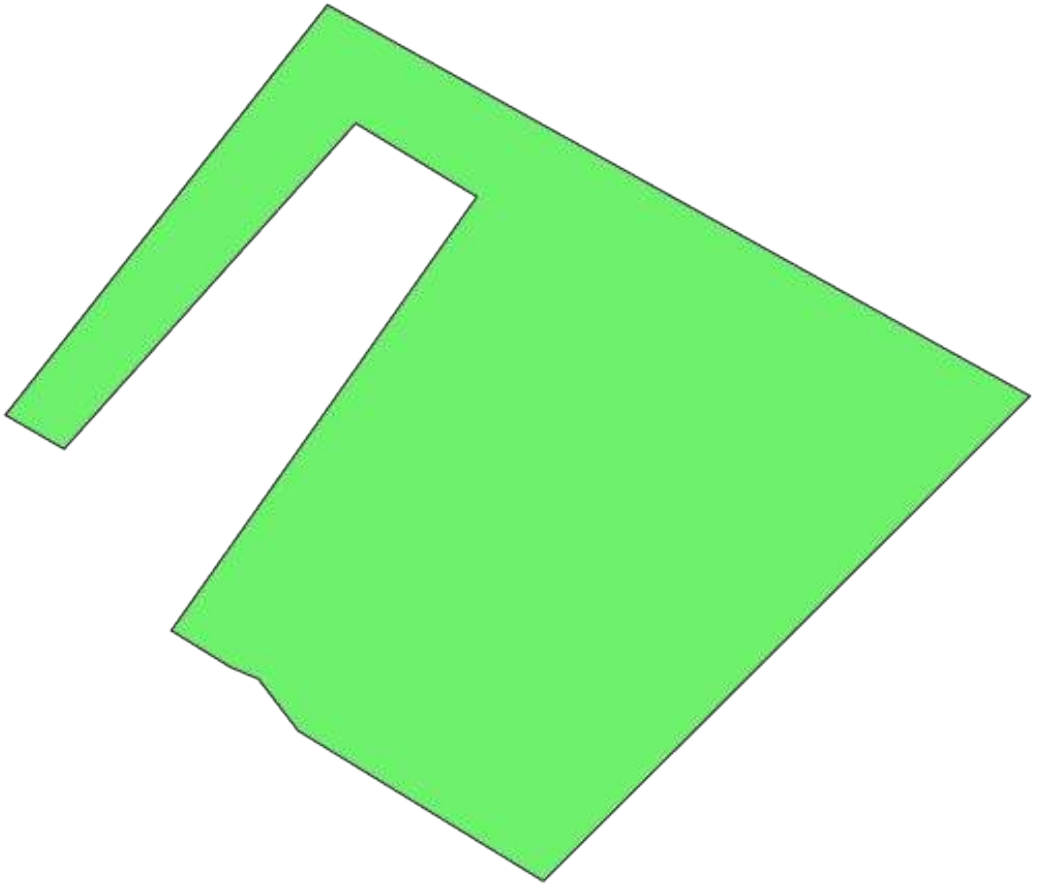


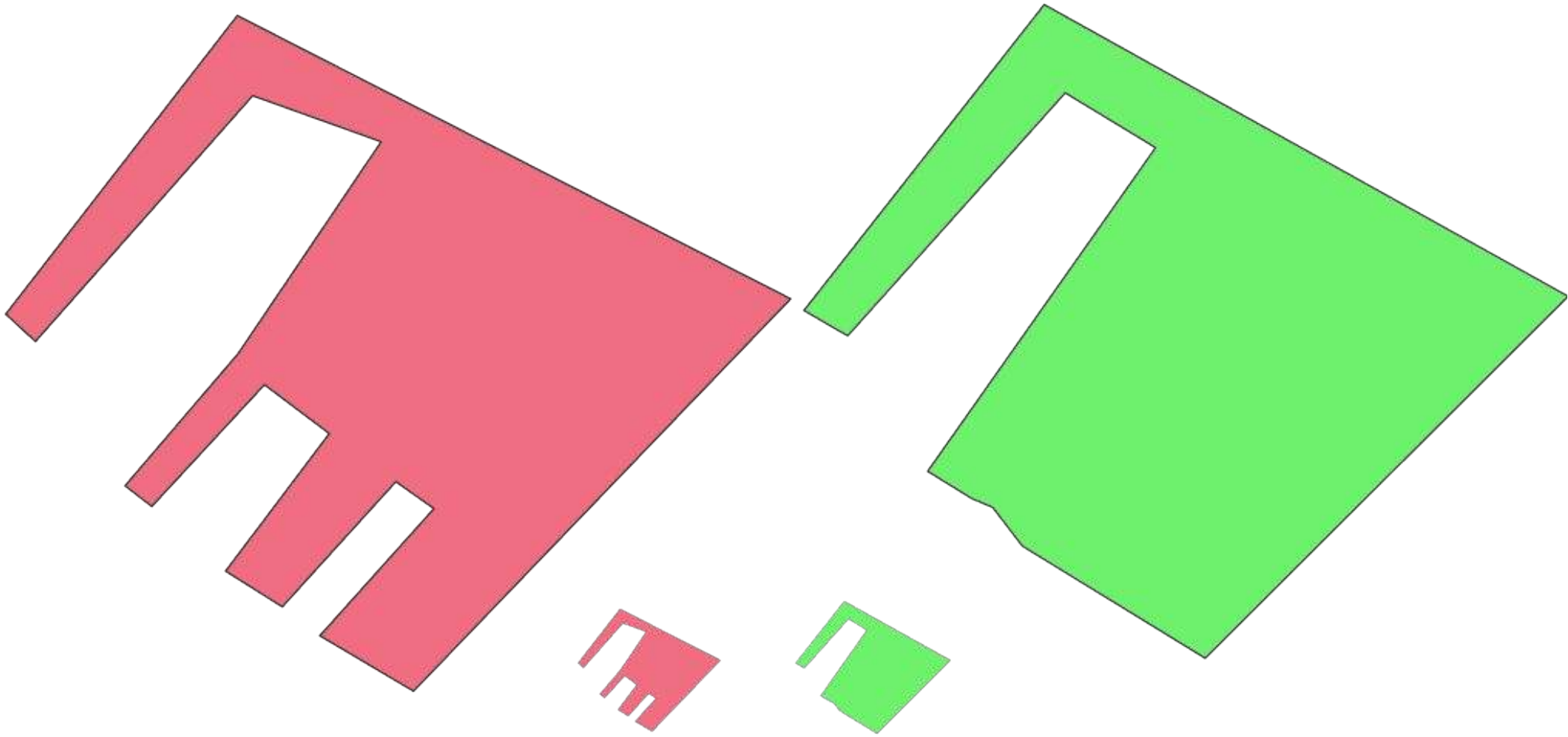
SELECT **ST_Buffer(way, 10)** FROM ...

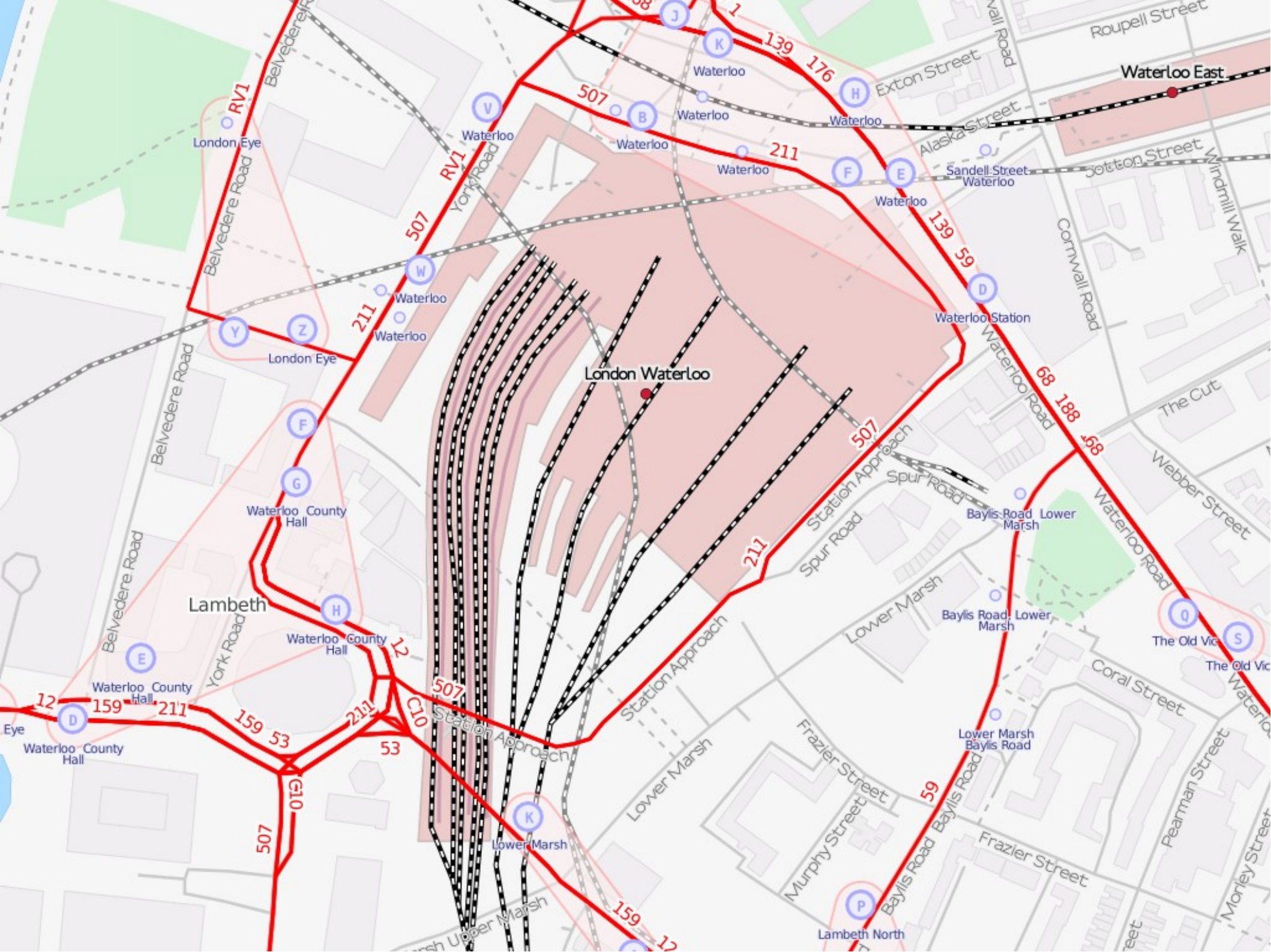


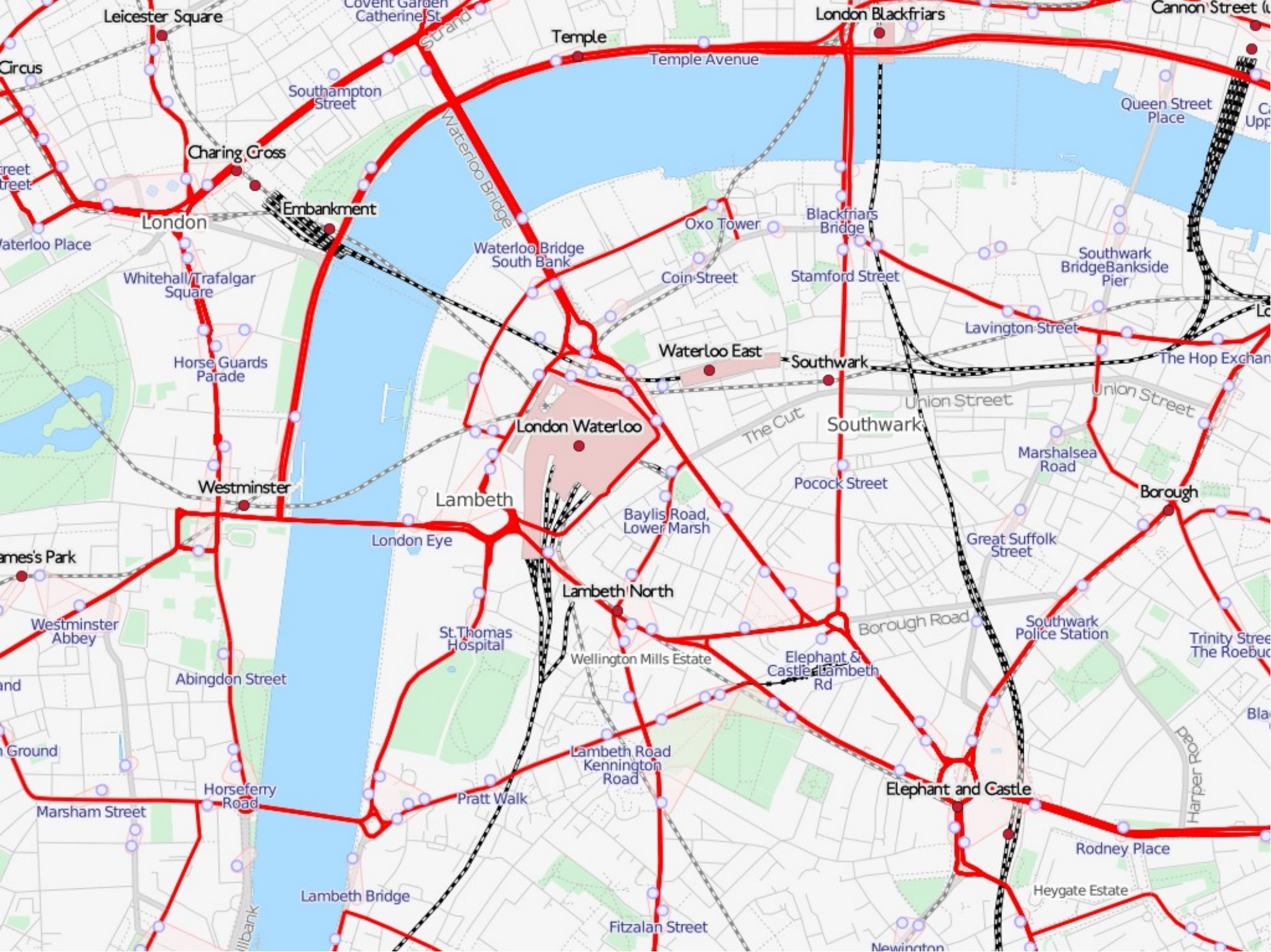


SELECT **ST_Buffer(**ST_Buffer(way, 10)**, -10)** FROM ...

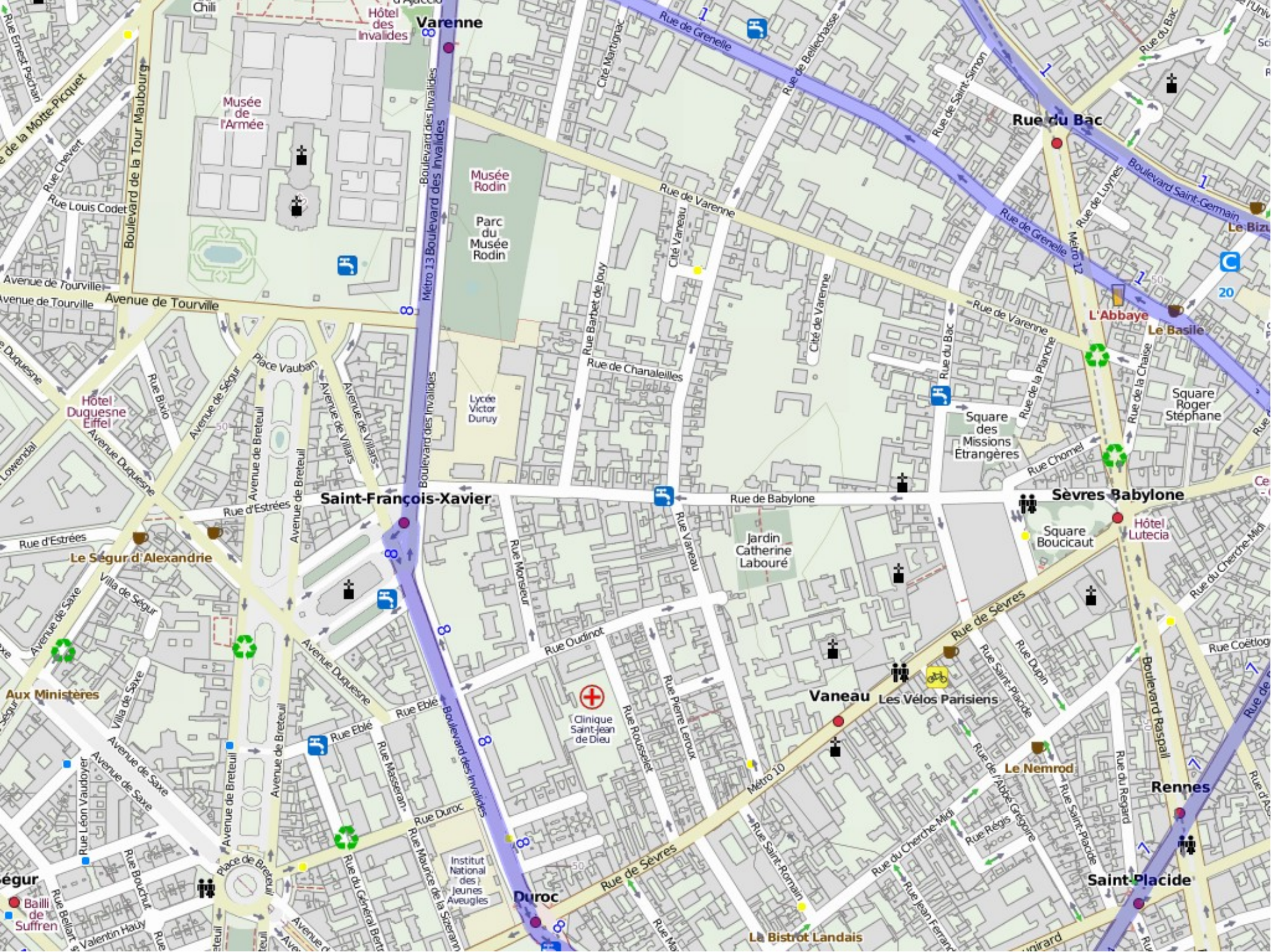





London Waterloo
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York Road
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Coral Street
Frazier Street
Murphy Street
Pearman Street
Morley Street
Lambeth North
Lower Marsh Baylis Road
Baylis Road Lower Marsh
Waterloo County Hall
London Eye
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G10

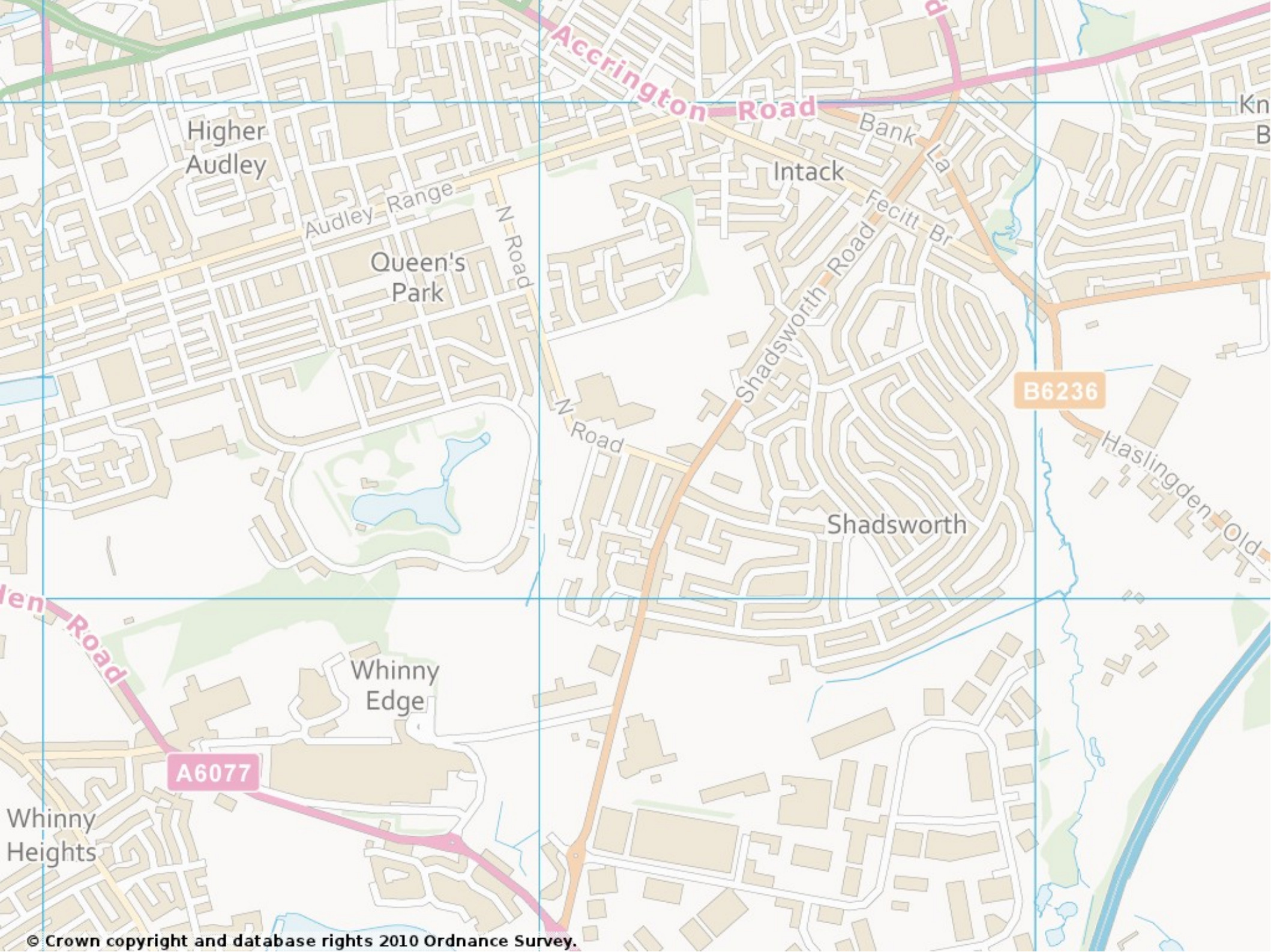

Leicester Square
Covent Garden
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Temple Avenue
London Blackfriars
Cannon Street
Circus
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Charing Cross
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London
Oxo Tower
Blackfriars Bridge
Waterloo Bridge South Bank
Coin Street
Stamford Street
Southwark Bridge
Bankside Pier
Whitehall
Trafalgar Square
Lavington Street
Waterloo East
Southwark
Horse Guards Parade
Union Street
The Cut
London Waterloo
Southwark
Marshalsea Road
Westminster
Lambeth
Pocock Street
Borough
Baylis Road, Lower Marsh
London Eye
Great Suffolk Street
Lambeth North
Westminster Abbey
St Thomas Hospital
Borough Road
Southwark Police Station
Wellington Mills Estate
Elephant & Castle
Lambeth Rd
Abingdon Street
Lambeth Road
Kennington Road
Horseferry Road
Pratt Walk
Elephant and Castle
Harper Road
Marsham Street
Rodney Place
Lambeth Bridge
Heygate Estate
Fitzalan Street

Merging Adjacent Polygons and stuff


Varenne
Rue du Bac
Saint-François-Xavier
Sèvres Babylone
Duroc
Vaneau
Saint-Placide
Rennes
Le Nemrod
Le Bistrot Landais
Aux Ministères
Le Ségur d'Alexandrie
Le Basile
L'Abbaye
Le Bizu
Hôtel des Invalides
Musée de l'Armée
Musée Rodin
Parc du Musée Rodin
Lycée Victor Duruy
Jardin Catherine Labouré
Square des Missions Étrangères
Square Roger Stéphane
Square Boucicaut
Hôtel Lutecia
Hôtel Duguesne Eiffel
Clinique Saint-Jean de Dieu
Institut National des Jeunes Aveugles
Les Vélos Parisiens
Boulevard des Invalides
Métro 13
Métro 12
Métro 10
Rue de Varenne
Rue de Grenelle
Boulevard Saint-Germain
Rue de Babylone
Rue de Sèvres
Rue du Cherche-Midi
Avenue de Tourville
Avenue de Breteuil
Avenue de Saxe
Avenue Duquesne
Rue d'Estrées
Rue Oudinot
Rue Duroc
Rue Eblé
Rue de Chanaleilles
Rue Barbet de Jouy
Rue Vaneau
Rue Monsieur
Rue Chomel
Rue du Regard
Rue Dupin
Rue Saint-Placide
Rue Saint-Romain
Rue de l'Abbé Grégoire
Rue Régis
Rue Pierre Leroux
Rue Rousselet
Rue Masseran
Rue Maurice de la Sizeranne
Rue du Général Bertholet
Rue Léon Vaudoyer
Rue Bouchut
Rue Belliard
Rue Valentin Haüy
Rue Louis Codet
Rue Chevert
Rue Ernest Psichari
Rue de la Motte-Picquet
Boulevard de la Tour Maubourg
Rue de Bellechasse
Rue de Saint-Simon
Cité Martignac
Cité Varneau
Cité de Varenne
Rue de Luynes
Rue de la Planche
Rue de la Chaise
Rue du Bac
Rue de Lowendal
Rue Bixio
Avenue de Ségur
Avenue de Villars
Place Vauban
Villa de Ségur
Place de Breteuil
Métro 12
Métro 10
Métro 8
Métro 1
Métro 7
Métro 9
RER C
20
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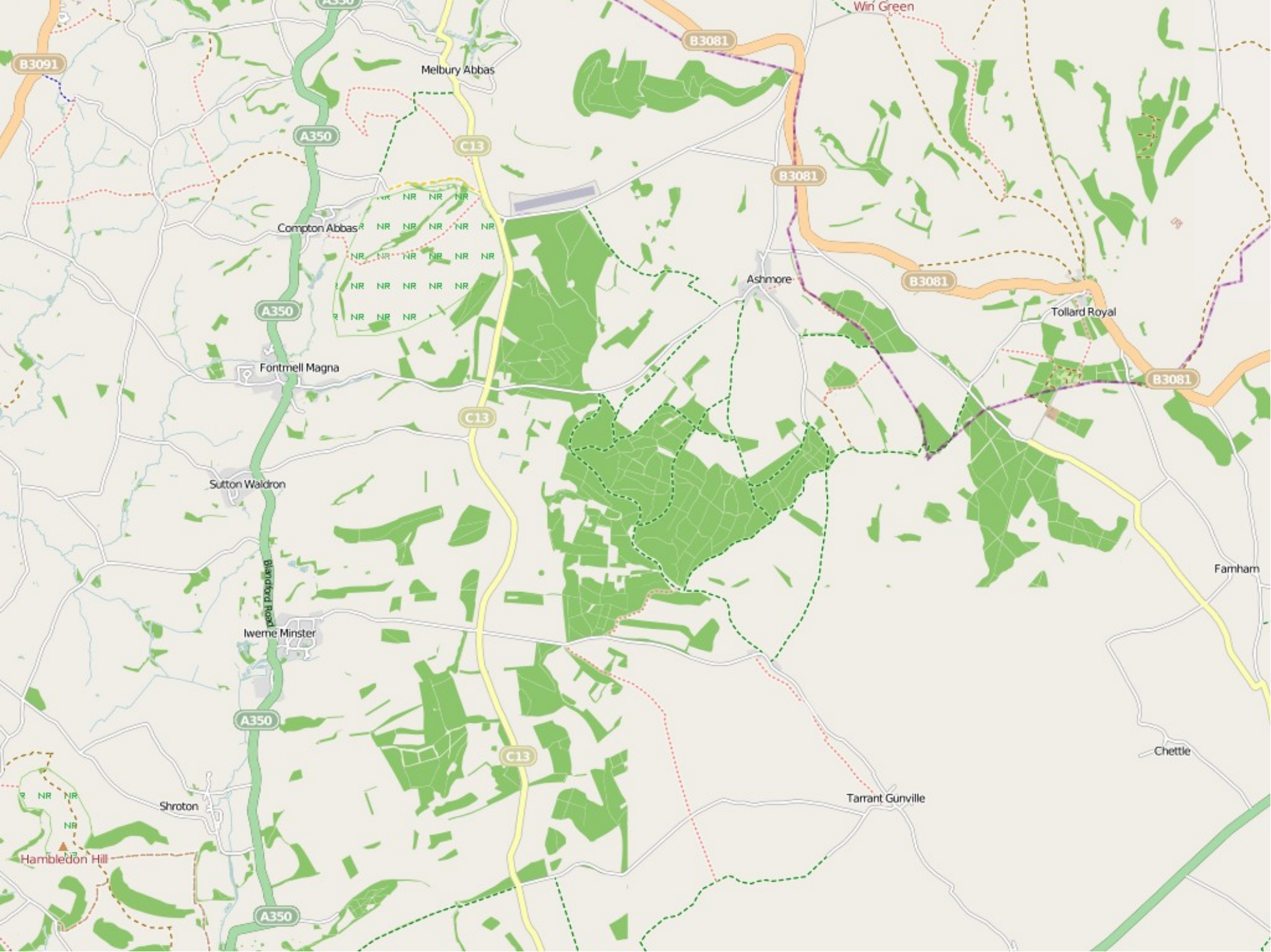
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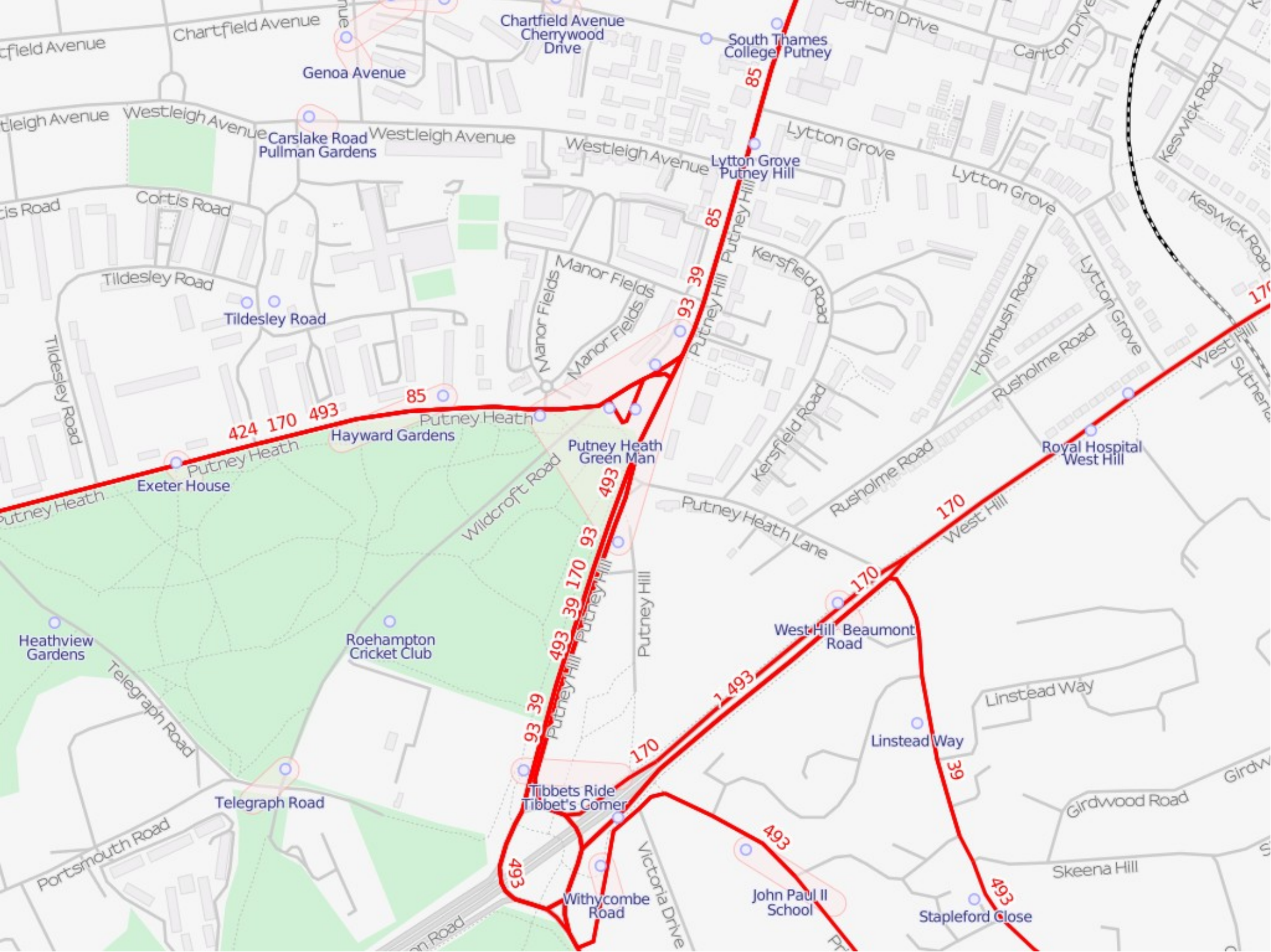
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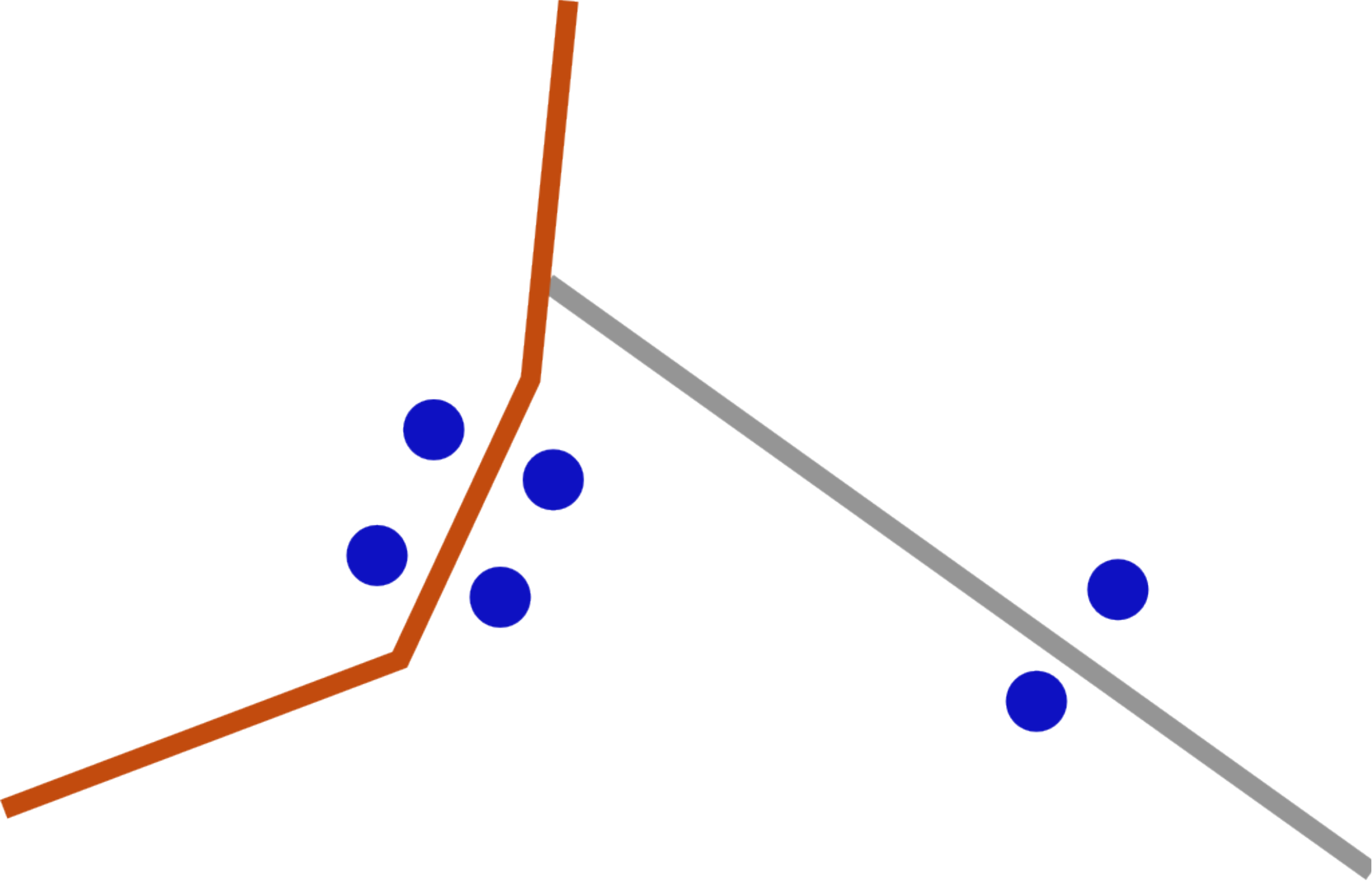
Dealing with import artefacts




Wiri Green
B3081
B3091
Melbury Abbas
A350
C13
B3081
Compton Abbas
NR
Ashmore
B3081
A350
Tollard Royal
Fontmell Magna
B3081
C13
Sutton Waldron
Blandford Road
Iwerne Minster
Farnham
A350
C13
Chettle
Tarrant Gunville
Shroton
NR
Hambledon Hill
A350

bus stop zones


Chartfield Avenue
Chartfield Avenue
Cherrywood Drive
Genoa Avenue
South Thames College Putney
Carlton Drive
Carlton Drive
Keswick Road
Westleigh Avenue
Carslake Road
Pullman Gardens
Westleigh Avenue
Westleigh Avenue
Lytton Grove
Lytton Grove
Putney Hill
Lytton Grove
Keswick Road
Cortis Road
Tildesley Road
Tildesley Road
Manor Fields
Manor Fields
Putney Hill
Kersfield Road
Holmbush Road
Lytton Grove
Tildesley Road
Rusholme Road
West Hill
424
170
493
85
Putney Heath
Hayward Gardens
Putney Heath
Exeter House
Putney Heath
Putney Heath
Green Man
Wildcroft Road
Kersfield Road
Rusholme Road
Royal Hospital
West Hill
493
Putney Heath Lane
170
West Hill
93
170
39
493
Putney Hill
Putney Hill
Putney Hill
170
West Hill
Beaumont Road
Heathview Gardens
Roehampton Cricket Club
Telegraph Road
1
493
Linstead Way
Linstead Way
93
39
Putney Hill
170
39
Tibbets Ride
Tibbet's Corner
Telegraph Road
Girdwood Road
493
Portsmouth Road
Victoria Drive
Skeena Hill
493
493
Withycombe Road
John Paul II School
Stapleford Close
493



SELECT way FROM planet_osm_point
WHERE highway = 'bus_stop'



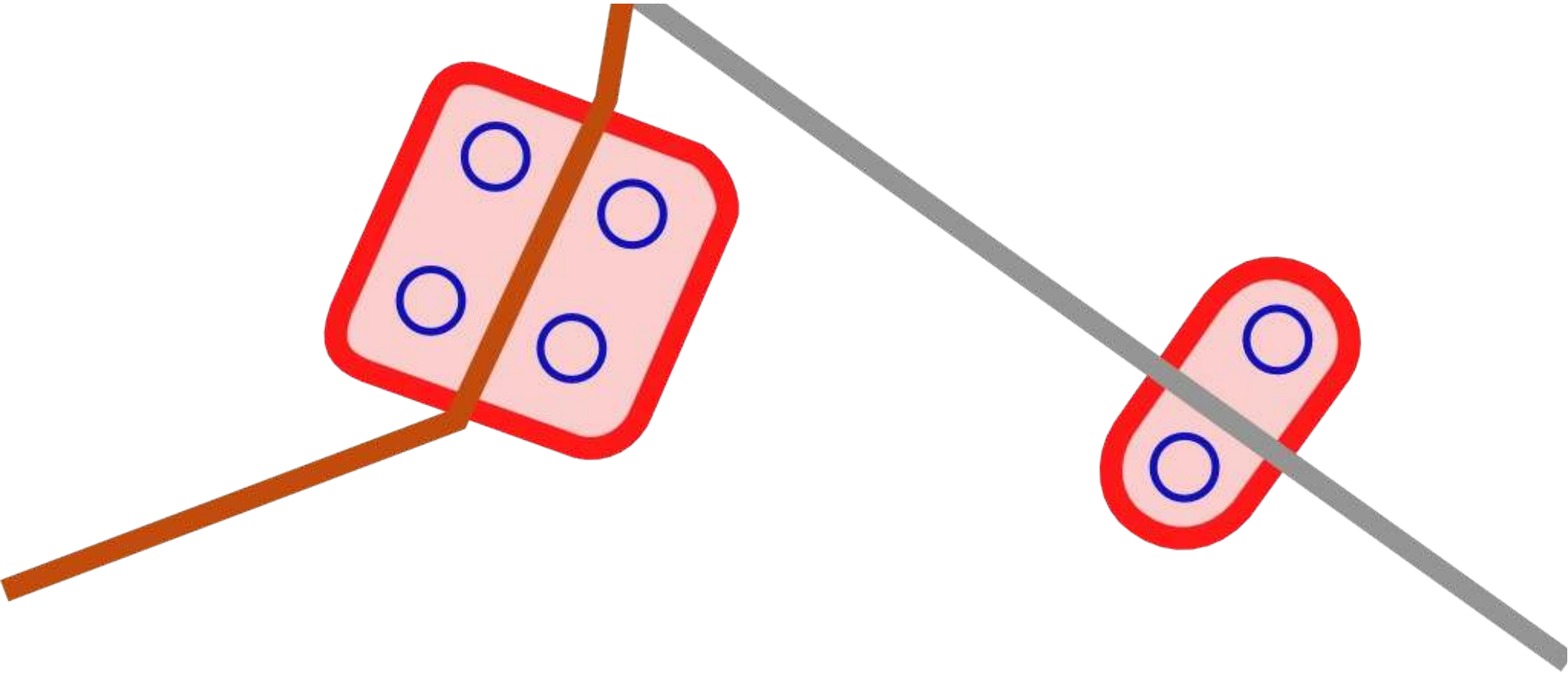
SELECT **ST_Collect(**way**)** FROM planet_osm_point
WHERE highway = 'bus_stop'
GROUP BY name

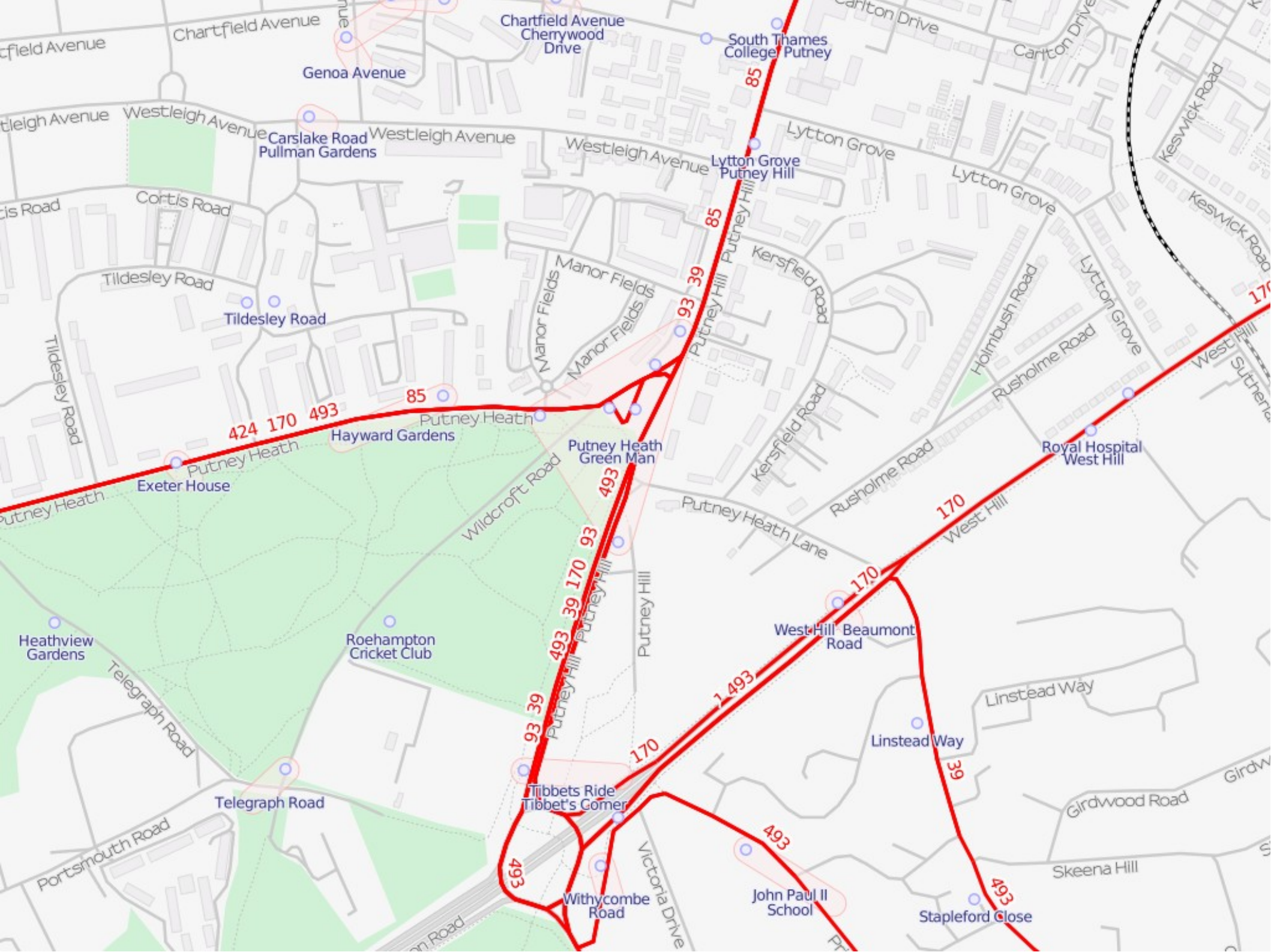


SELECT **ST_ConvexHull(**ST_Collect(way)**)**
FROM planet_osm_point
WHERE highway = 'bus_stop'
GROUP BY name



```
SELECT ST_Buffer(ST_ConvexHull(ST_Collect(way)),25)
FROM planet_osm_point
WHERE highway = 'bus_stop'
GROUP BY name
```



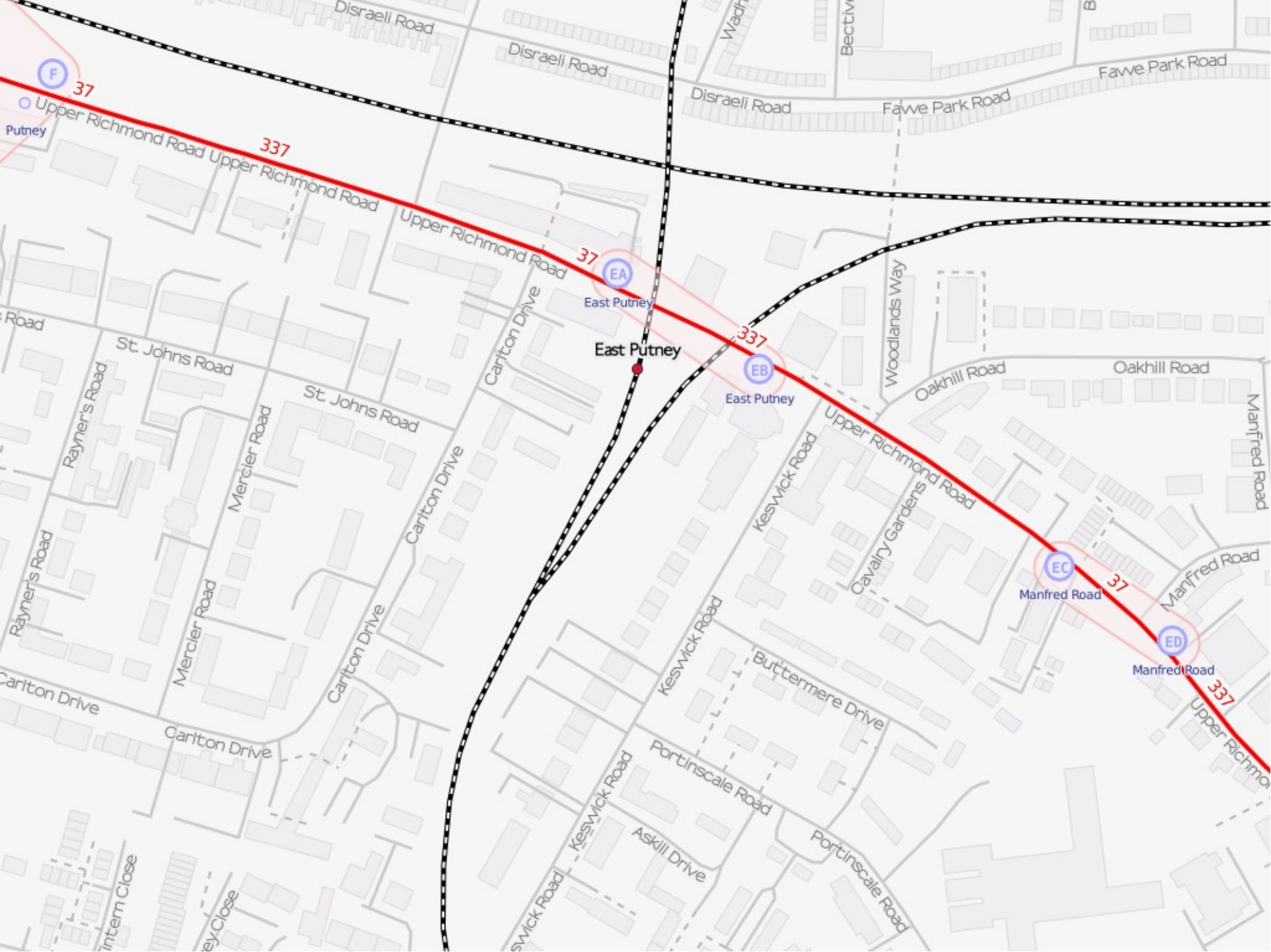

Chartfield Avenue
Chartfield Avenue
Cherrywood Drive
Genoa Avenue
South Thames College Putney
Carlton Drive
Carlton Drive
Keswick Road
Westleigh Avenue
Carslake Road
Pullman Gardens
Westleigh Avenue
Westleigh Avenue
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Putney Hill
Lytton Grove
Keswick Road
Cortis Road
Tildesley Road
Tildesley Road
Manor Fields
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Kersfield Road
Holmbush Road
Lytton Grove
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85
Putney Heath
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Putney Heath
Putney Heath
Green Man
Wildcroft Road
Kersfield Road
Rusholme Road
Royal Hospital
West Hill
493
Putney Heath Lane
170
West Hill
93
170
39
493
Putney Hill
Putney Hill
Putney Hill
170
West Hill
Beaumont Road
Heathview Gardens
Roehampton Cricket Club
Telegraph Road
1
493
Linstead Way
Linstead Way
93
39
Putney Hill
170
39
Tibbets Ride
Tibbet's Corner
Telegraph Road
Girdwood Road
493
Portsmouth Road
Victoria Drive
Skeena Hill
493
493
Withycombe Road
John Paul II School
Stapleford Close
493

Convex Hull vs. Concave Hull

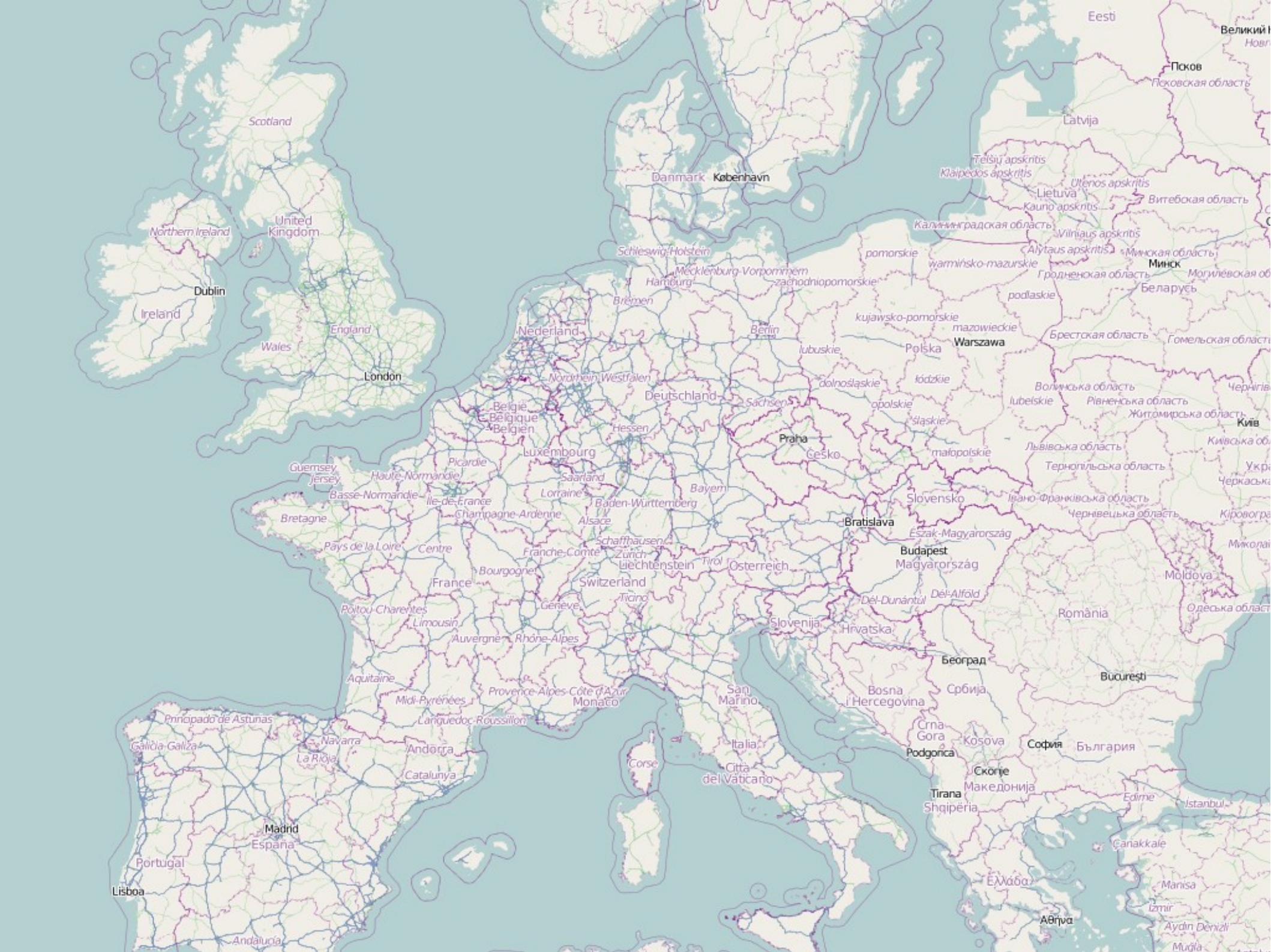


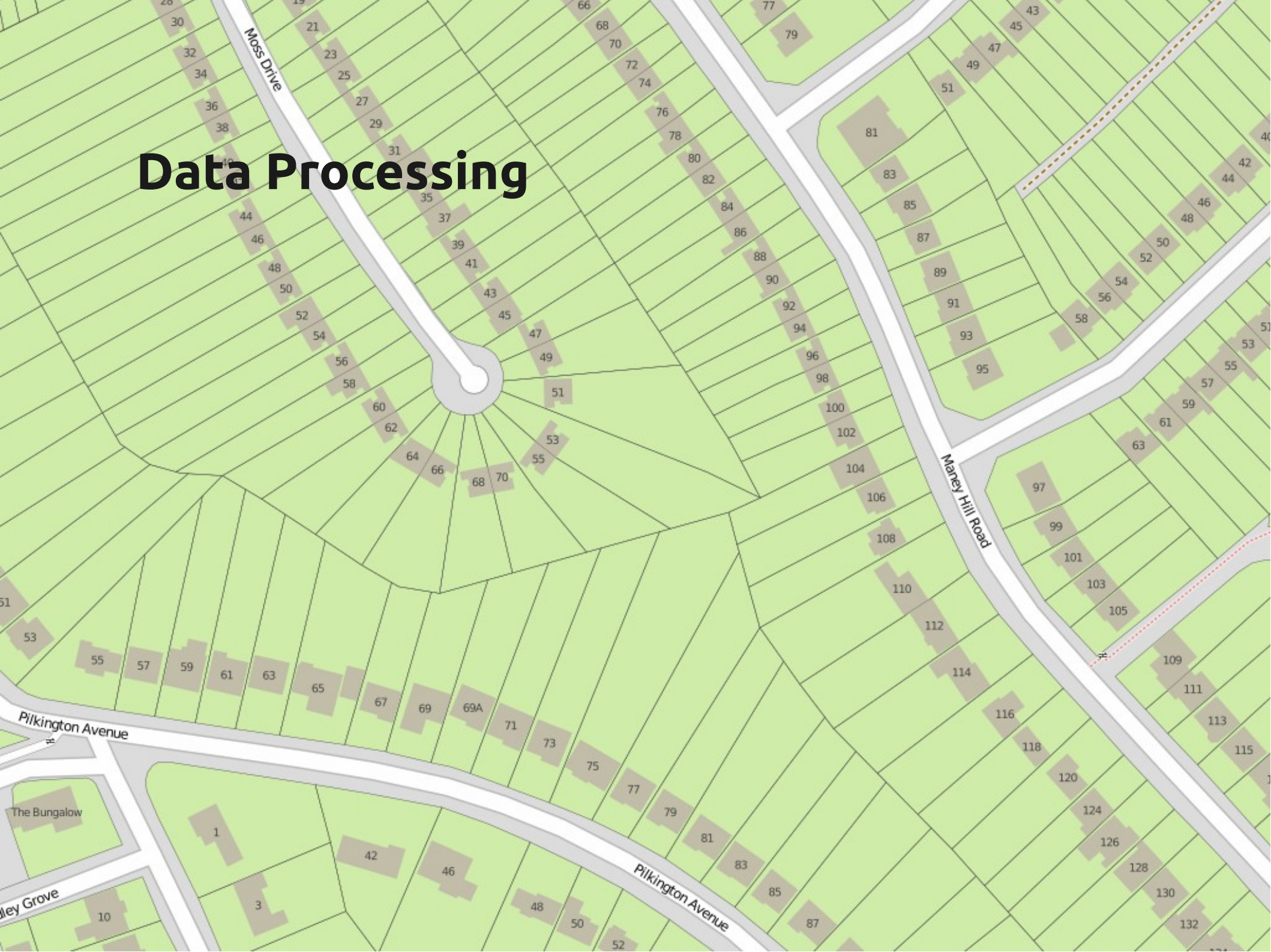
http://www.bostongis.com/postgis_concavehull.snippet

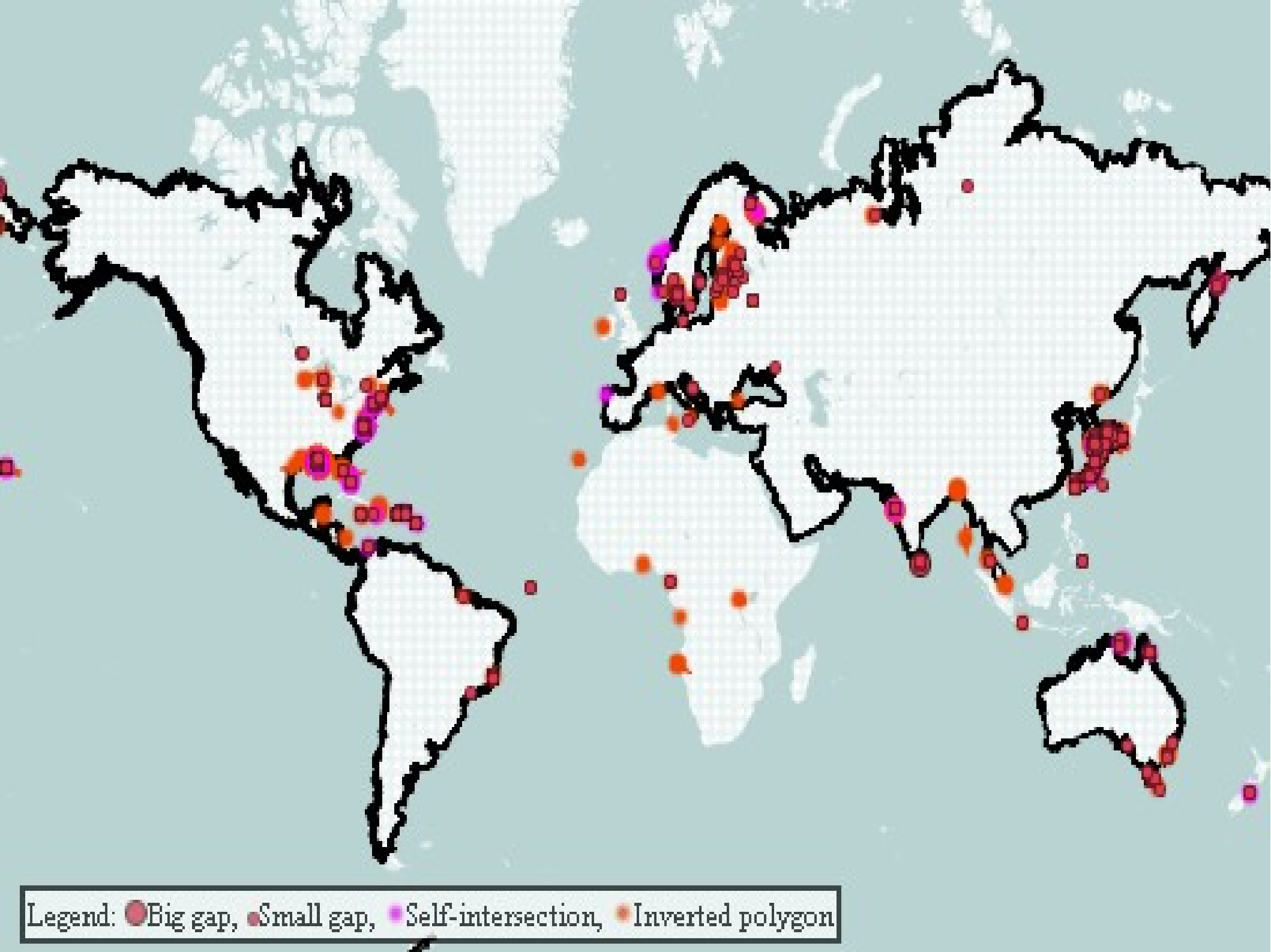
bus routes names


Disraeli Road
Disraeli Road
Disraeli Road
Fawe Park Road
Fawe Park Road
F
37
Upper Richmond Road
Putney
337
Upper Richmond Road
Upper Richmond Road
37
EA
East Putney
East Putney
337
EB
East Putney
Woodlands Way
Oakhill Road
Oakhill Road
Manfred Road
Upper Richmond Road
Keswick Road
Cavalry Gardens
EC
Manfred Road
37
Manfred Road
ED
Manfred Road
337
Upper Richmond
St. Johns Road
St. Johns Road
Rayner's Road
Mercier Road
Carlton Drive
Carlton Drive
Rayner's Road
Mercier Road
Carlton Drive
Carlton Drive
Carlton Drive
Keswick Road
Buttermere Drive
Portinscale Road
Keswick Road
Askill Drive
Portinscale Road
Keswick Road

Squiggles hate dashes and transparency


Eesti
Великий Н
Псков
Псковская область
Latvija
Scotland
Danmark
København
Telšių apskritis
Klaipėdos apskritis
Utenos apskritis
Lietuva
Витебская область
Kauno apskritis
Калининградская область
Vilniaus apskritis
Alytaus apskritis
Минская область
Минск
Northern Ireland
United Kingdom
Schleswig-Holstein
pomorskie
warmińsko-mazurskie
Гродненская область
Могилевская об
Mecklenburg-Vorpommern
Hamburg
zachodniopomorskie
Беларусь
Dublin
Ireland
Bremen
podlaskie
kujawsko-pomorskie
mazowieckie
Брестская область
Гомельская область
Nederland
Berlin
England
Wales
lubuskie
Polska
Warszawa
London
Nordrhein-Westfalen
łódzkie
Волинська область
Чернігів
dolnośląskie
Deutschland
Sachsen
lubelskie
Рівненська область
Житомирська область
België
Belgique
Belgien
opolskie
śląskie
Київ
Hessen
Praha
Česko
małopolskie
Київська об
Luxembourg
Львівська область
Guernsey
Jersey
Picardie
Haute-Normandie
Saarland
Тернопільська область
Basse-Normandie
Lorraine
Bayern
Slovensko
Івано-Франківська область
Ile-de-France
Baden-Württemberg
Champagne-Ardenne
Чернівецька область
Bretagne
Alsace
Bratislava
Észak-Magyarország
Pays de la Loire
Centre
Schaffhausen
Franche-Comté
Zürich
Budapest
Liechtenstein
Tirol
Österreich
Magyarország
Bourgogne
France
Switzerland
Moldova
Ticino
Dél-Dunántúl
Dél-Alföld
Poitou-Charentes
Genève
Limousin
Slovenija
Hrvatska
România
Auvergne
Rhône-Alpes
Београд
Aquitaine
Србија
Bosna
i Hercegovina
Provence-Alpes-Côte d'Azur
Monaco
San
Marino
Midi-Pyrénées
Languedoc-Roussillon
Principado de Asturias
Crna
Gora
Kosova
София
България
Galicia-Galiza
Navarra
Andorra
La Rioja
Italia
Podgorica
Corse
Catalunya
Città
del Vaticano
Скопје
Македонија
Tirana
Shqipëria
Madrid
España
Portugal
Lisboa
Ελλάδα
Αθήνα
Andalucía


Data Processing
Moss Drive
Maney Hill Road
Pilkington Avenue
Pilkington Avenue
The Bungalow


Legend: Big gap, Small gap, Self-intersection, Inverted polygon

Dotspotting

Dots on the pavement from flickr by ericrodenbeck created on July 8, 2011 at 02:43 PM



FILTER TABLE:

Dot	Coordinates	flickr:id	tags	ownername	datetaken	description	woeid	accuracy	tit
206733	37.768499,-122.417167	5810413500	blue dot	dotspotting	2011-05-13 20:01:51	—	55970976	16	IMG_2
206735	37.765166,-122.418834	5809848809	blue dot	dotspotting	2011-05-11 12:58:11	—	23512048	16	IMG_2



[map=yes] mapquest

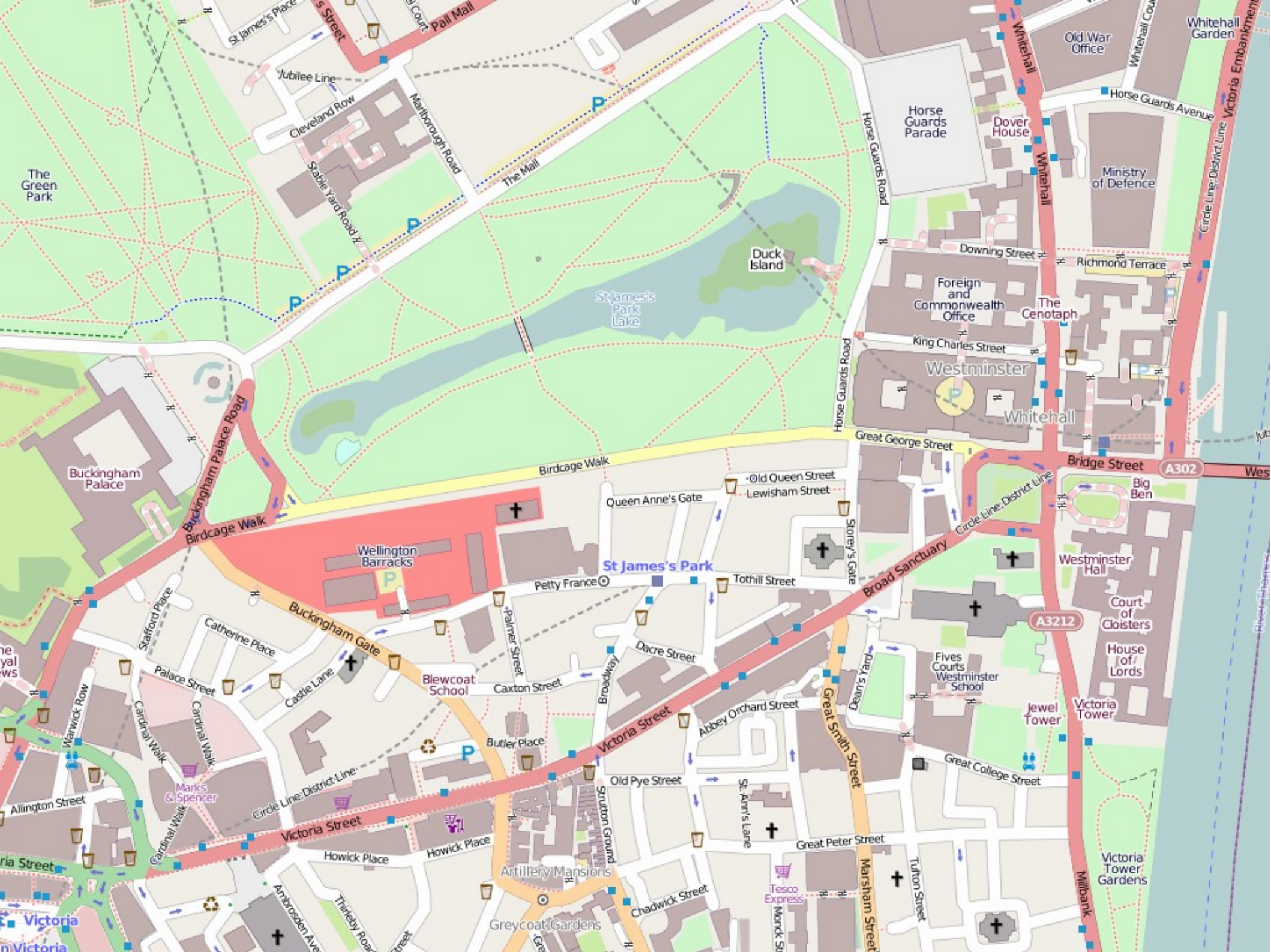
A new kind of map based on open data



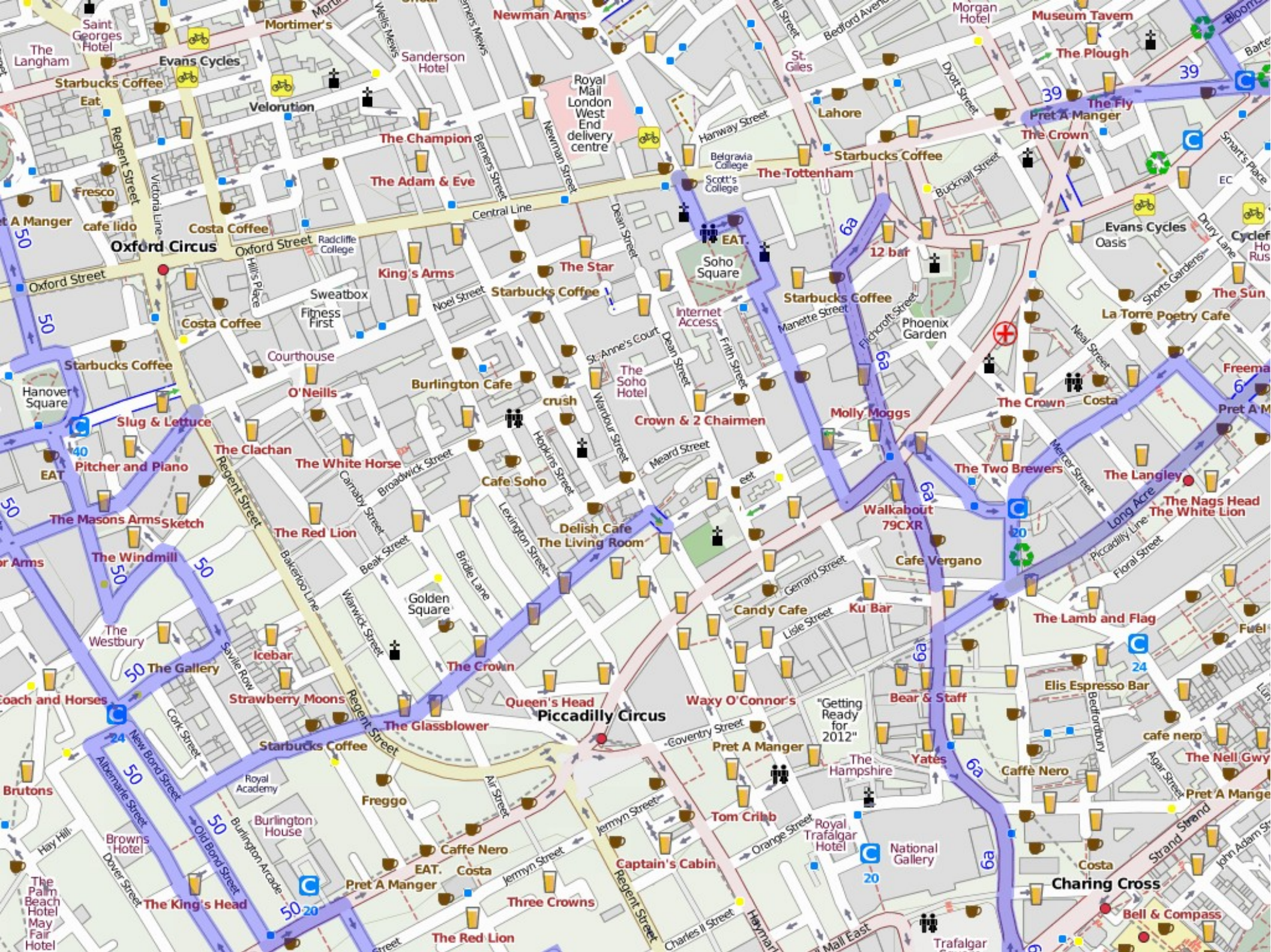
Most online maps are are designed to help you get around in a car. This generally means displaying: roads, businesses, buildings, on-ramps, parks, oceans and traffic congestion. Nothing wrong with that! Designers get handed a tool kit that has as many tools as a good swiss army knife, and the maps reflect these tools. Millions of people use them to make appointments across town, find restaurants, and drive home for the holidays.

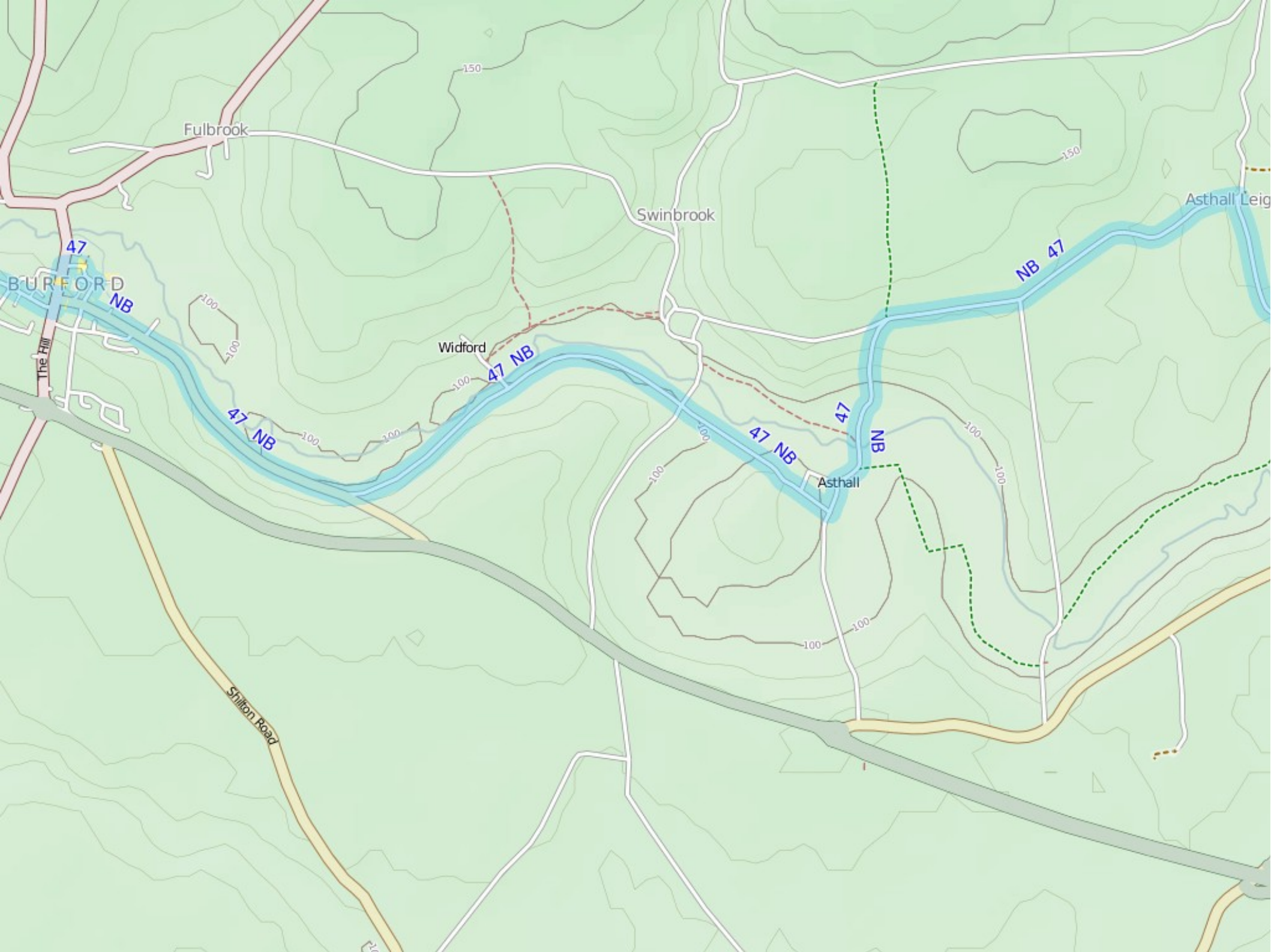
But what if, instead of a swiss army knife, we used a box of crayons? Or charcoal and newsprint? Or play-doh? What would those maps look like? What could they tell us about the world?



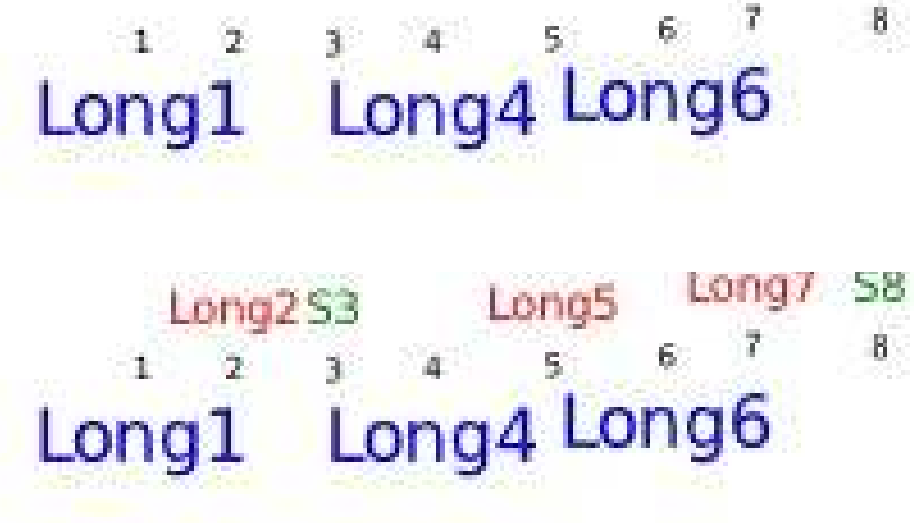

St James's Place
Pall Mall
Jubilee Line
Cleveland Row
Marlborough Road
The Mall
Stable Yard Road
The Green Park
Duck Island
St James's Park Lake
Horse Guards Road
Horse Guards Parade
Dover House
Whitehall
Old War Office
Whitehall Garden
Horse Guards Avenue
Victoria Embankment
Circle Line; District Line
Ministry of Defence
Downing Street
Richmond Terrace
Foreign and Commonwealth Office
The Cenotaph
King Charles Street
Westminster
Whitehall
Great George Street
Bridge Street
A302
Big Ben
Buckingham Palace
Buckingham Palace Road
Birdcage Walk
Old Queen Street
Lewisham Street
Queen Anne's Gate
Storey's Gate
Broad Sanctuary
Westminster Hall
Wellington Barracks
St James's Park
Petty France
Tothill Street
Court of Cloisters
A3212
Buckingham Gate
Stafford Place
Catherine Place
Palmer Street
Dacre Street
Broadway
Palace Street
Castle Lane
Blewcoat School
Caxton Street
Fives Courts Westminster School
House of Lords
Warwick Row
Cardinal Walk
Abbey Orchard Street
Jewel Tower
Victoria Tower
Butler Place
Victoria Street
Great College Street
Old Pye Street
Allington Street
Mark's & Spencer
Howick Place
Strutton Ground
St Ann's Lane
Great Peter Street
Great Smith Street
Marsham Street
Tufton Street
Millbank
Victoria Tower Gardens
Artillery Mansions
Tesco Express
Chadwick Street
Greycoat Gardens
Victoria

POI Density


Oxford Circus
Piccadilly Circus
Charing Cross
Soho Square
Golden Square
Hanover Square
Oxford Street
Regent Street
Central Line
Victoria Line
Bakerloo Line
Piccadilly Line
Jermyn Street
Strand
Haymarket
Mall East
Trafalgar
Coventry Street
Orange Street
Air Street
Lisle Street
Gerrard Street
Meard Street
Broadwick Street
Beak Street
Noel Street
Hanway Street
Oxford Street
Dean Street
Newman Street
Berners Street
Wells Mews
Mortimer's
Saint Georges Hotel
The Langham
Evans Cycles
Starbucks Coffee
Eat
Velorution
Sanderson Hotel
Newman Arms
Royal Mail London West End delivery centre
The Champion
The Adam & Eve
Fresco
cafe lido
Costa Coffee
Radcliffe College
Hills Place
Sweatbox Fitness First
King's Arms
The Star
Starbucks Coffee
Internet Access
Courthouse
O'Neills
Burlington Cafe
crush
The Soho Hotel
Crown & 2 Chairmen
Slug & Lettuce
Pitcher and Piano
The Clachan
The White Horse
Cafe Soho
The Red Lion
Delish Cafe
The Living Room
The Masons Arms
Sketch
The Windmill
The Westbury
The Gallery
Icebar
Strawberry Moons
The Glassblower
Queen's Head
The Crown
Waxy O'Connor's
Starbucks Coffee
Freggo
Caffe Nero
EAT.
Costa
Pret A Manger
Three Crowns
The Red Lion
Captain's Cabin
Tom Cribb
Royal Trafalgar Hotel
National Gallery
Yates
Pret A Manger
The Hampshire
"Getting Ready for 2012"
Bear & Staff
Caffè Nero
Elis Espresso Bar
Costa
Bell & Compass
Walkabout 79CXR
Cafe Vergano
Candy Cafe
Ku Bar
The Two Brewers
Molly Moggs
The Crown
Costa
The Lamb and Flag
The Nags Head
The White Lion
The Langley
Phoenix Garden
La Torre Poetry Cafe
The Sun
12 bar
Starbucks Coffee
Evans Cycles
Oasis
Museum Tavern
The Plough
Pret A Manger
The Fly
The Crown
Lahore
Starbucks Coffee
The Tottenham
Belgravia College
Scott's College
Royal Academy
Burlington House
Burlington Arcade
Albemarle Street
New Bond Street
Old Bond Street
Cork Street
Savile Row
Hay Hill
Dover Street
Brutons
Browns Hotel
The Palm Beach Hotel
May Fair Hotel
The King's Head
Coach and Horses
Starbucks Coffee
Eat
Pret A Manger
Fresco
cafe lido
Costa Coffee
Starbucks Coffee
Courthouse
Hanover Square
Slug & Lettuce
Pitcher and Piano
EAT
The Masons Arms
Sketch
The Windmill
The Westbury
The Gallery
Coach and Horses
Brutons
Starbucks Coffee
Royal Academy
Burlington House
Browns Hotel
Hay Hill
Dover Street
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May Fair Hotel
The King's Head
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Pret A Manger
The Fly
The Crown
Lahore
Starbucks Coffee
The Tottenham
Belgravia College
Scott's College
Soho Square
EAT.
Internet Access
Starbucks Coffee
The Star
King's Arms
Sweatbox Fitness First
Costa Coffee
Courthouse
O'Neills
Burlington Cafe
crush
The Soho Hotel
Crown & 2 Chairmen
The Clachan
The White Horse
Cafe Soho
The Red Lion
Delish Cafe
The Living Room
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Strawberry Moons
The Glassblower
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The Crown
Waxy O'Connor's
Piccadilly Circus
Golden Square
Regent Street
Oxford Street
Central Line
Victoria Line
Bakerloo Line
Piccadilly Line
Jermyn Street
Strand
Haymarket
Mall East
Trafalgar
Coventry Street
Orange Street
Air Street
Lisle Street
Gerrard Street
Meard Street
Broadwick Street
Beak Street
Noel Street
Hanway Street
Oxford Street
Dean Street
Newman Street
Berners Street
Wells Mews
Mortimer's
Saint Georges Hotel
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Royal Mail London West End delivery centre
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cafe lido
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Radcliffe College
Hills Place
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King's Arms
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Starbucks Coffee
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Courthouse
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Crown & 2 Chairmen
Slug & Lettuce
Pitcher and Piano
The Clachan
The White Horse
Cafe Soho
The Red Lion
Delish Cafe
The Living Room
The Masons Arms
Sketch
The Windmill
The Westbury


Fulbrook
Swinbrook
Asthall Leig
BURFORD
The Hill
Widford
Asthall
Shilton Road
47 NB
NB 47
150
100

Text Processing



East Birmingham Street
East Birmingham St
E Birmingham St
E. B'ham St
E. B'ham

Skeletons

Rule 2: IF a branch that leaves from a junction point does not leave the circle around its origin THEN eliminate it.

Applying these rules finally reduces the skeleton to the centrelines (see figure 8).



Figure 6. Aggregation of parcels declared as roads in the ALK

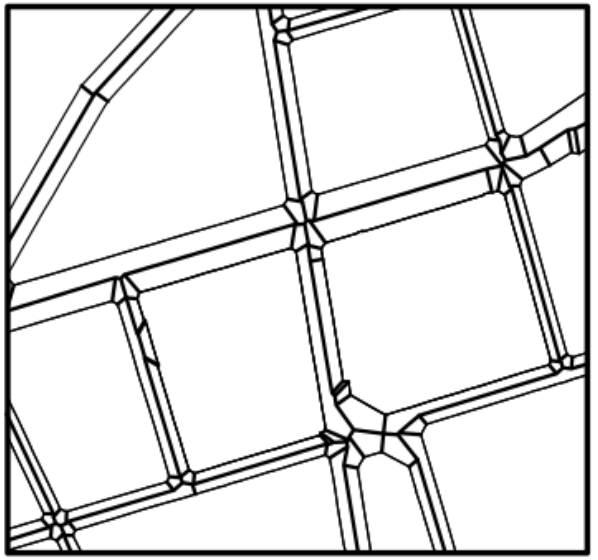


Figure 7. Straight Skeleton of aggregated road parcels

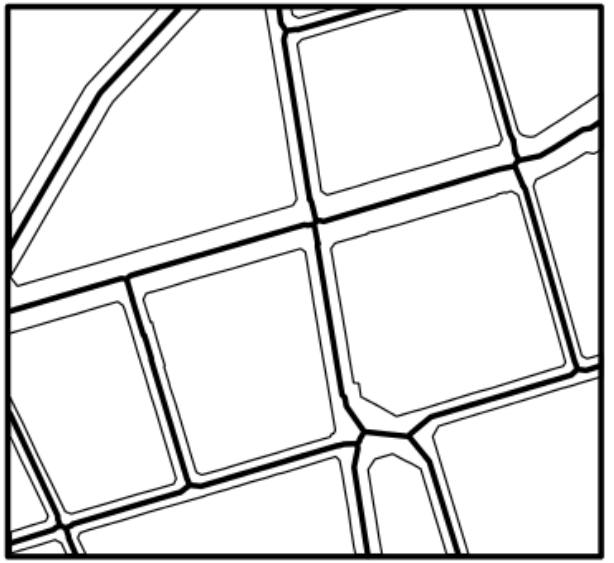


Figure 8. Centrelines of polygon

Still these centrelines differ from the optimal solution, since a junction does not appear as an intersection point of road axes. Instead of that, a centreline enters the surrounding of a junction in a rather unnatural arc, because road parcels widen in these areas. Also, at T-junctions, the straight line is bent into the direction of the incoming road. To solve this problem an algorithm has been developed which is able to change most of the junctions to the desired appearance. This algorithm mainly relies on rules which define when a junction point has to be repositioned and in which cases multiple junction points can be replaced by a single point. These rules are essentially based on a definition of junction areas. Basically, the new locations of the junction points are

migurski / Skeletron

Watch | Fork | 3 | 1

Computes straight skeletons of simple polygons — Read more
http://skeletron.teczno.com:8206

Downloads

HTTP | Git Read-Only | https://github.com/migurski/Skeletron.git | Read-Only access

Wrote a linemerge() that works



Michal Migurski (author)
February 20, 2011

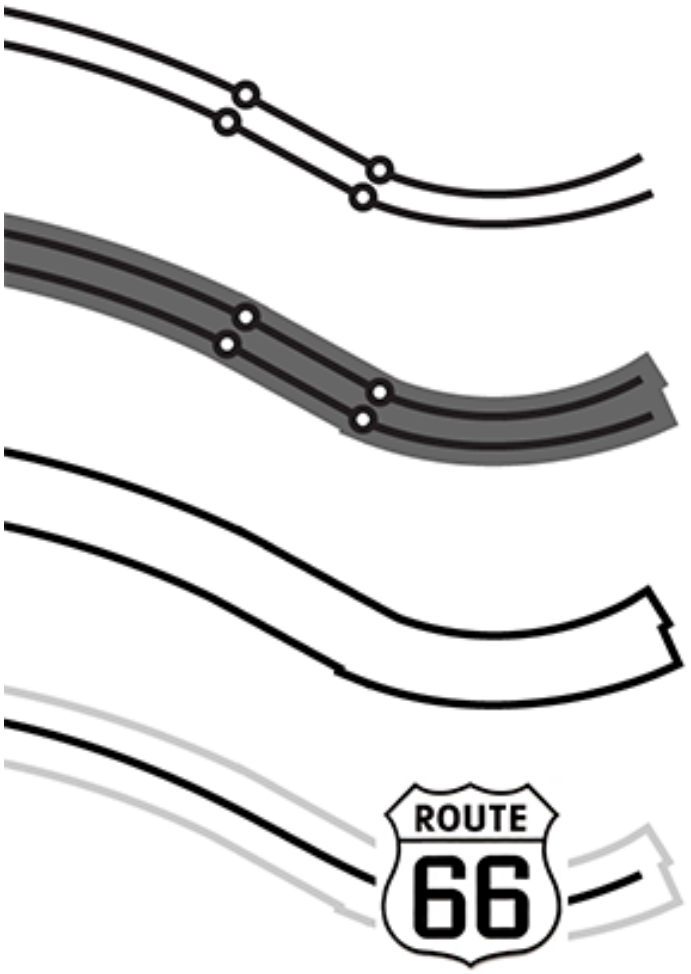
commit c0e2699d9a98c5fd4ed9
tree 63f20b04fd5c056fa02a
parent 3f12a432423ce2af7f81

Skeletron /

name	age	message	history
CGAL/	February 20, 2011	Wrote a linemerge() that works [Michal Migurski]	
Skeletron/	December 27, 2010	Dropped threshold for tiny angles still further to... [Michal Migurski]	
.gitignore	December 19, 2010	Made old sketch-skeleton into more-proper Skeletro... [Michal Migurski]	
README.txt	December 19, 2010	Tiny readme [Michal Migurski]	
skeletron-test.py	December 27, 2010	... [Michal Migurski]	

README.txt

The straight skeleton:
http://twak.blogspot.com/2009/01/that-straight-skeleton-again.html


ROUTE
66


Thank you!
Moss Drive
Maney Hill Road
Pilkington Avenue
Pilkington Avenue
The Bungalow